



คณะแพทยศาสตร์ มหาวิทยาลัยสงขลานครินทร์

Clinical correlation: SHOCK

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หน่วยเวชบำบัดวิกฤต

สาขาวิชาอายุรศาสตร์

คณะแพทยศาสตร์ มหาวิทยาลัยสงขลานครินทร์

Outlines

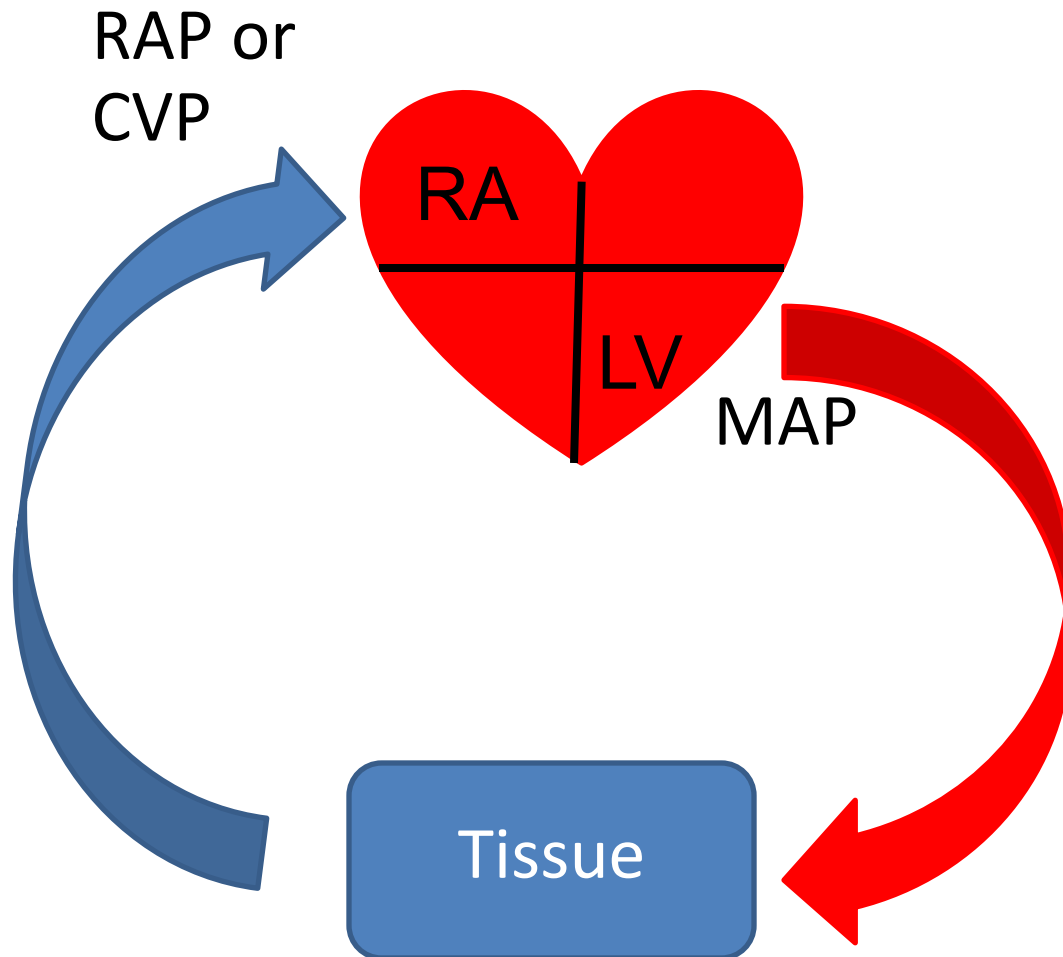
- Hemodynamic parameters
- Shock definition
- Shock classification
- Case study

SDL

- Venous return
- Cardiac arrest
- Drugs in cardiac arrest and emergency



Mean Arterial Pressure (MAP)

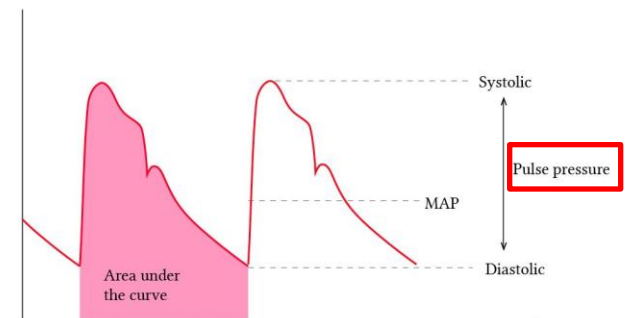


$$V = I \times R$$

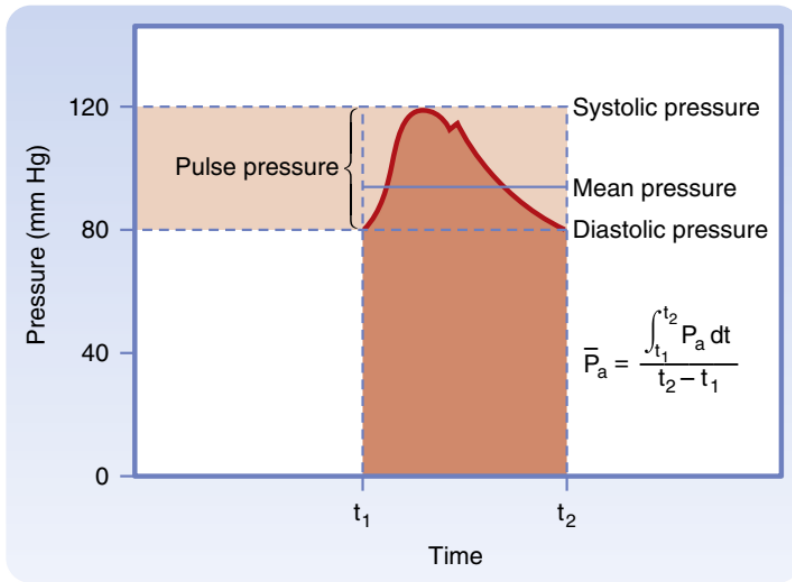
$$\text{MAP} - \text{CVP} = \text{CO} \times \text{SVR}$$

If $\text{CVP} = 0$;

$$\text{MAP} = \text{CO} \times \text{SVR}$$

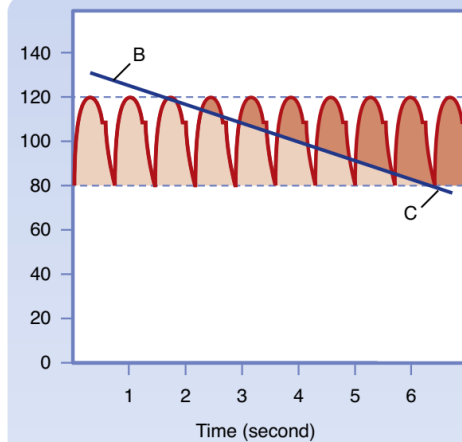


$$\begin{aligned} * \text{MAP} &= (\text{SBP} + 2\text{DBP})/3 \\ \text{MAP} &= \text{DBP} + 1/3 (\text{SBP} - \text{DBP}) \end{aligned}$$



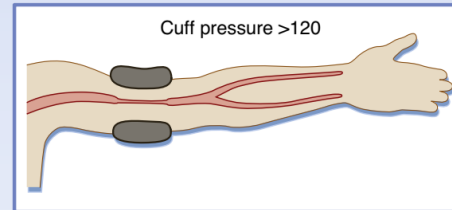
• **Fig. 17.13** Arterial systolic, diastolic, pulse, and mean pressure. Mean arterial pressure ($\bar{P}$) represents the area under the arterial pressure curve (dark red) divided by the duration of the cardiac cycle ($t_2 - t_1$).

$$\begin{aligned} *MAP &= (SBP + 2DBP)/3 \\ MAP &= DBP + 1/3 (SBP - DBP) \\ \text{MAP} &= DBP + 1/3 (PP) \end{aligned}$$



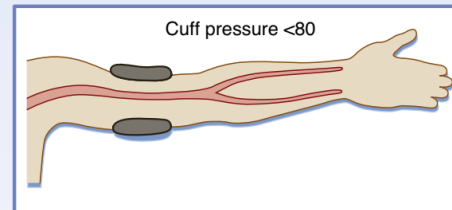
Consider that the arterial blood pressure is being measured in a patient whose blood pressure is 120/80 mm Hg. The pressure (represented by the oblique line) in a cuff around the patient's arm is allowed to fall from greater than 120 mm Hg (point B) to below 80 mm Hg (point C) in about 6 seconds.

A



When the cuff pressure exceeds the systolic arterial pressure (120 mm Hg), no blood progresses through the arterial segment under the cuff, and no sounds can be detected by a stethoscope bell placed on the arm distal to the cuff.

B

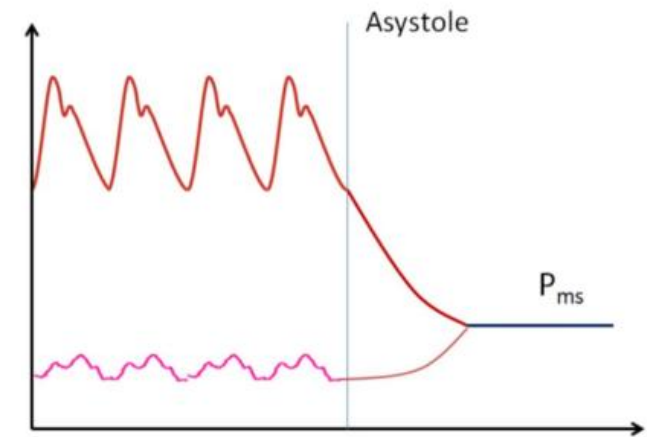
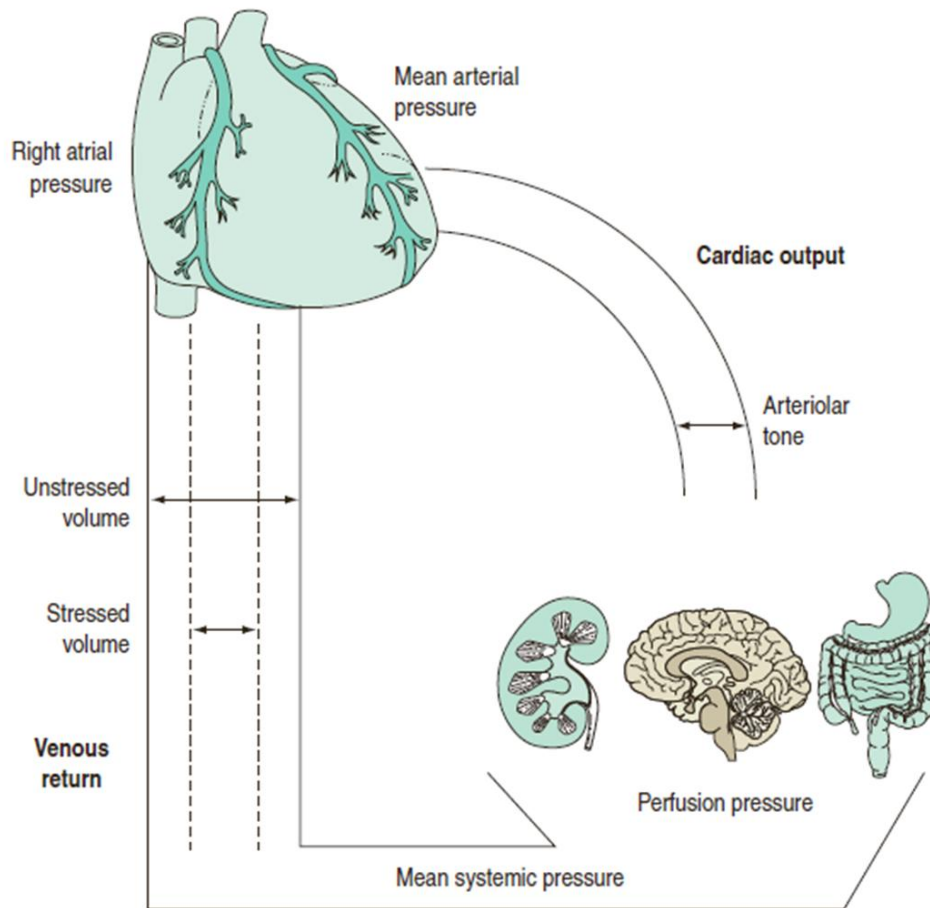


When the cuff pressure falls below the diastolic arterial pressure, arterial flow past the region of the cuff is continuous, and no sounds are audible. When the cuff pressure is between 120 and 80 mm Hg, spurts of blood traverse the artery segment under the cuff with each heartbeat, and the Korotkoff sounds are heard through the stethoscope.

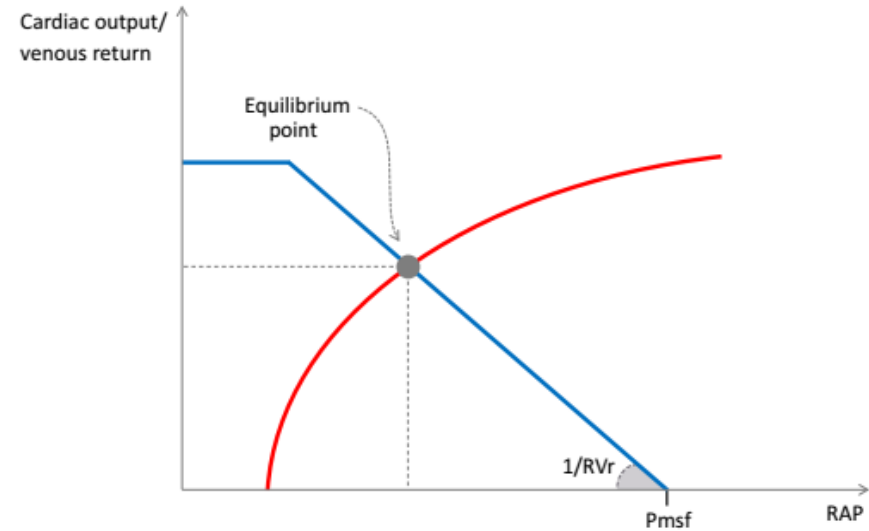
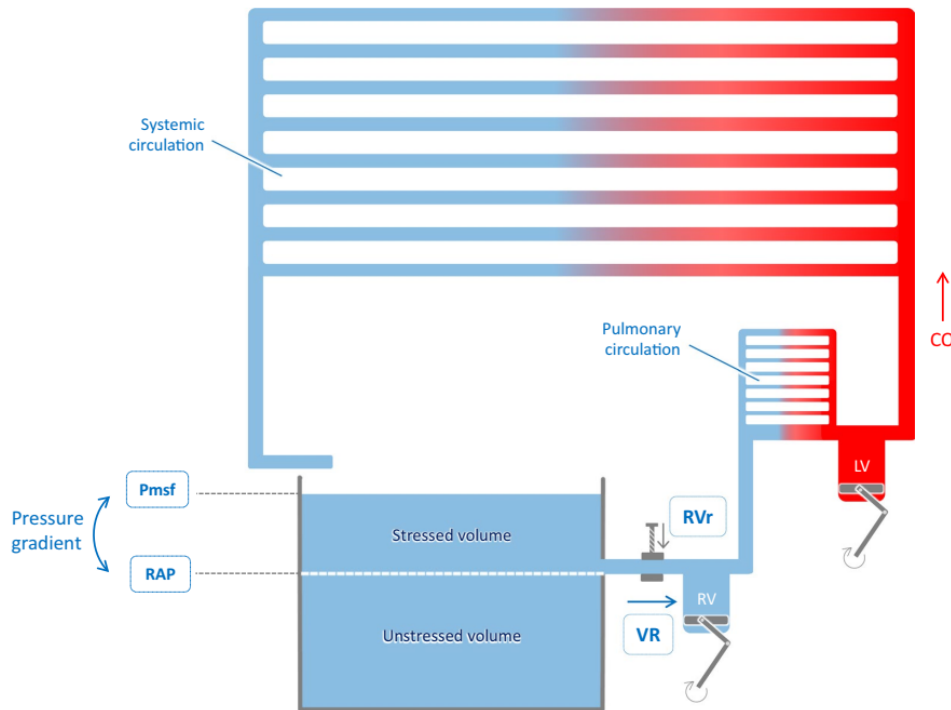
C

Mean systemic (filling) pressure (P_{ms} , P_{msf})

= the pressure that develops in the systemic circulation if the heart suddenly stops



Venous return



$$\text{Venous return} = (P_{\text{msf}} - \text{RAP}) / R_{\text{Vr}}$$

Hemodynamic formula

- $MAP - CVP = CO \times SVR$
 $= SV \times HR \times SVR$
 $= (EDV - ESV) \times HR \times SVR$
 $= \frac{EDV}{EDV} \times (EDV - ESV) \times HR \times SVR$
 $= EDV \times EF \times HR \times SVR$

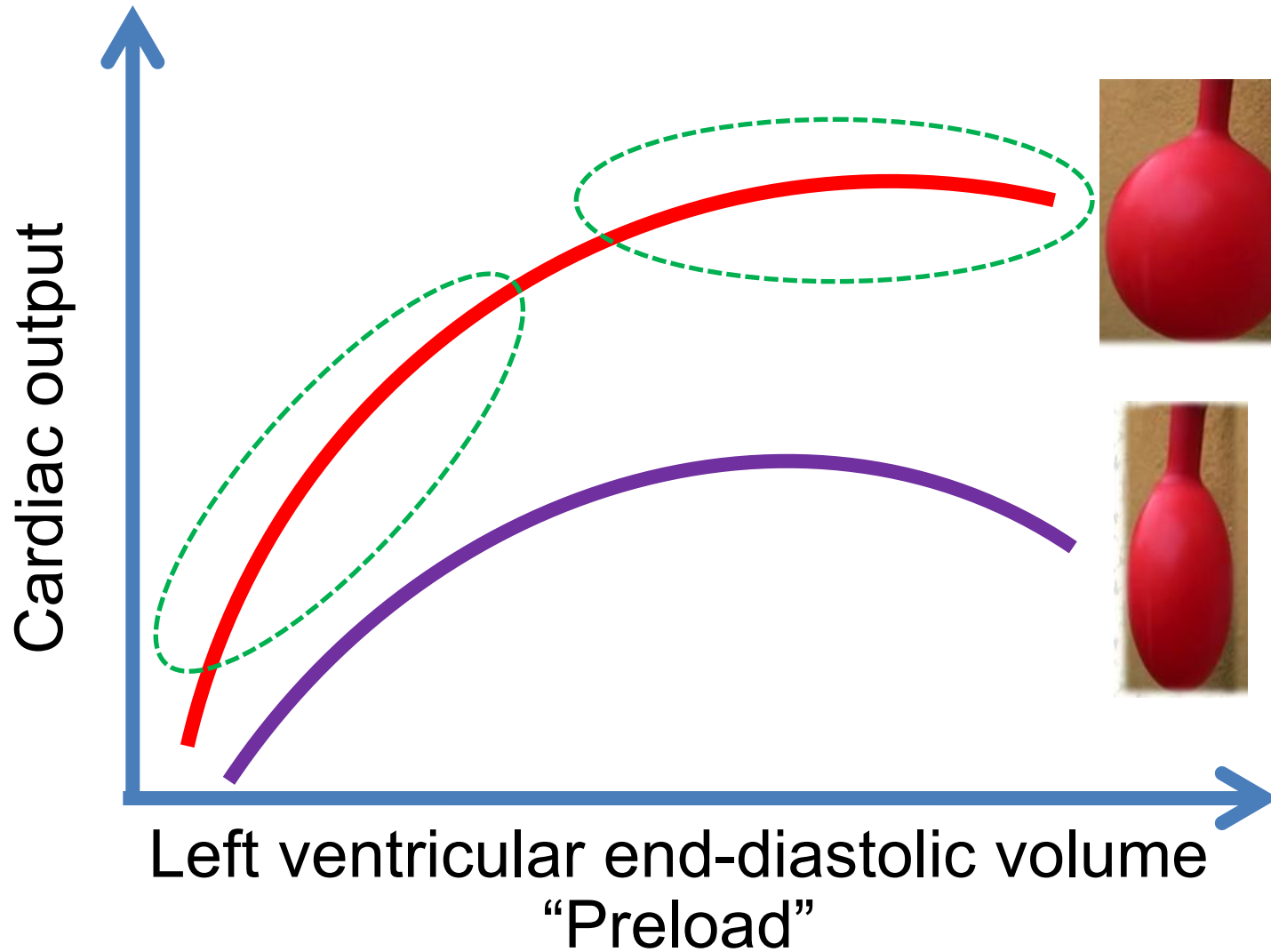
$$MAP - CVP = \text{preload} \times \text{contractility} \times HR \times \text{afterload}$$

Preload

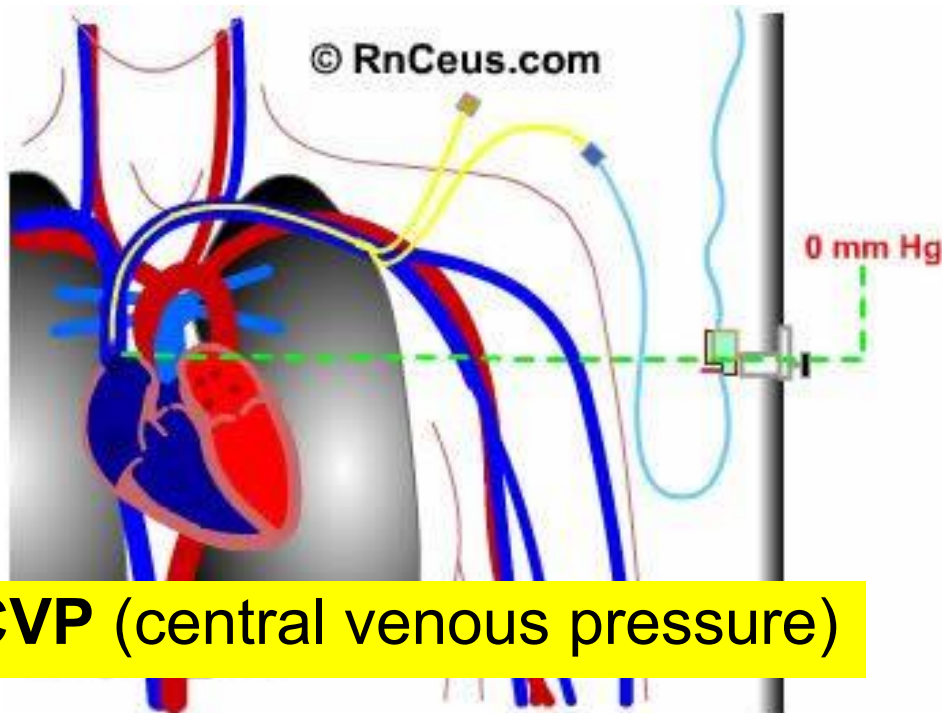
- Definitions
 - Amount of volume returning to the heart
 - Blood volume in the ventricle to be ejected



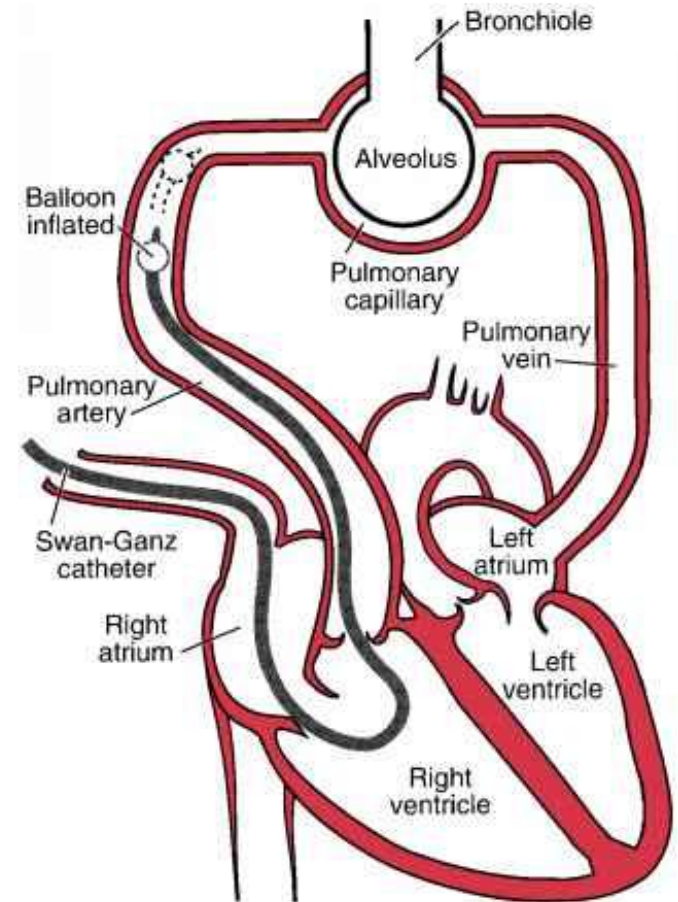
Frank-Starling law



How to measure preload?



CVP (central venous pressure)

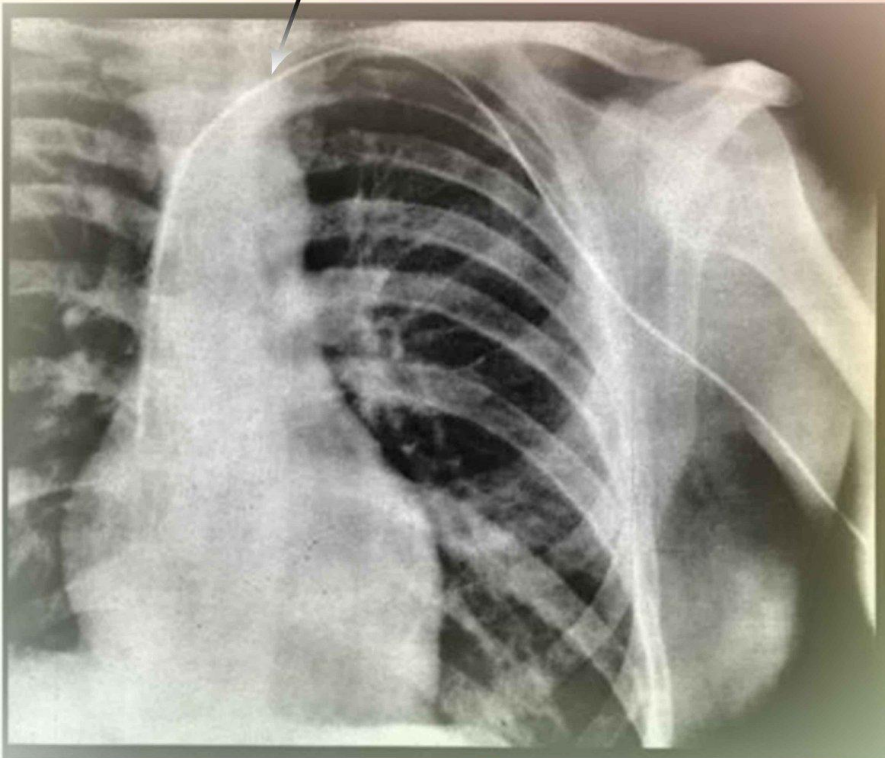


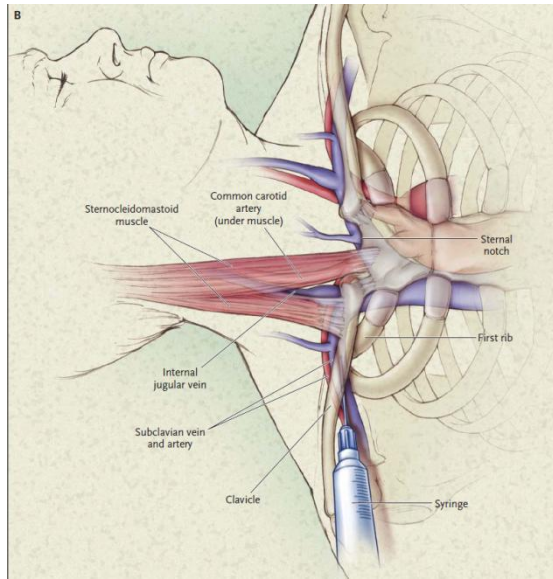
PCWP (pulmonary capillary wedge pressure)
PAOP (pulmonary artery occlusive pressure)

Central venous catheterization

In 1929, Werner Forssmann put himself under local anesthesia and inserted a catheter into a vein in his arm. Forssmann was successfully and safely passed the catheter into his heart.

It was very painful. I felt that I had planted an apple orchard and other men who had gathered the harvest stood at the wall, laughing at me,"

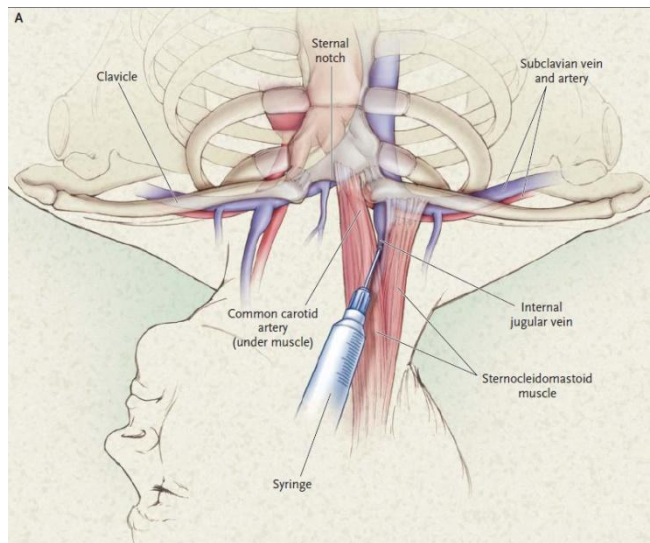




Subclavian vein

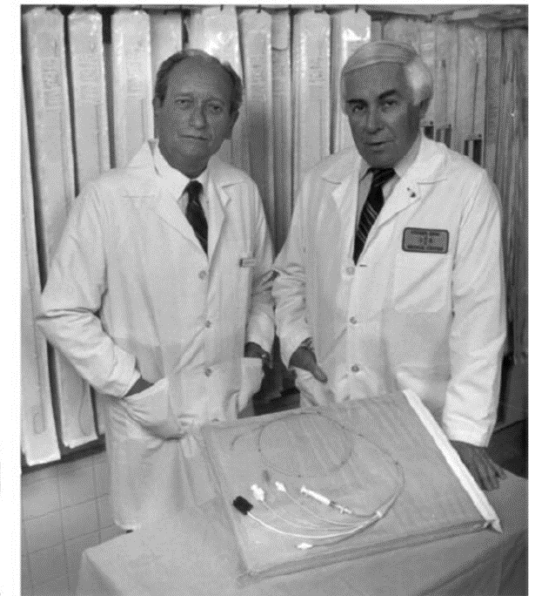
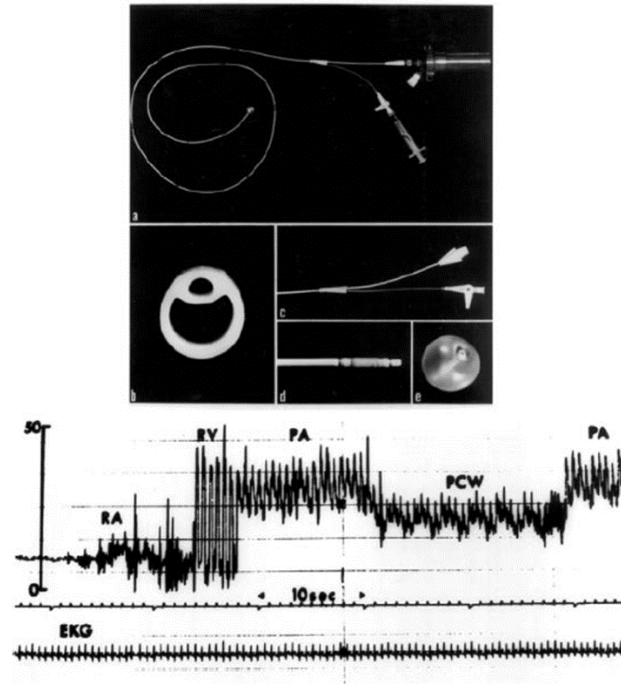


Venous cut down

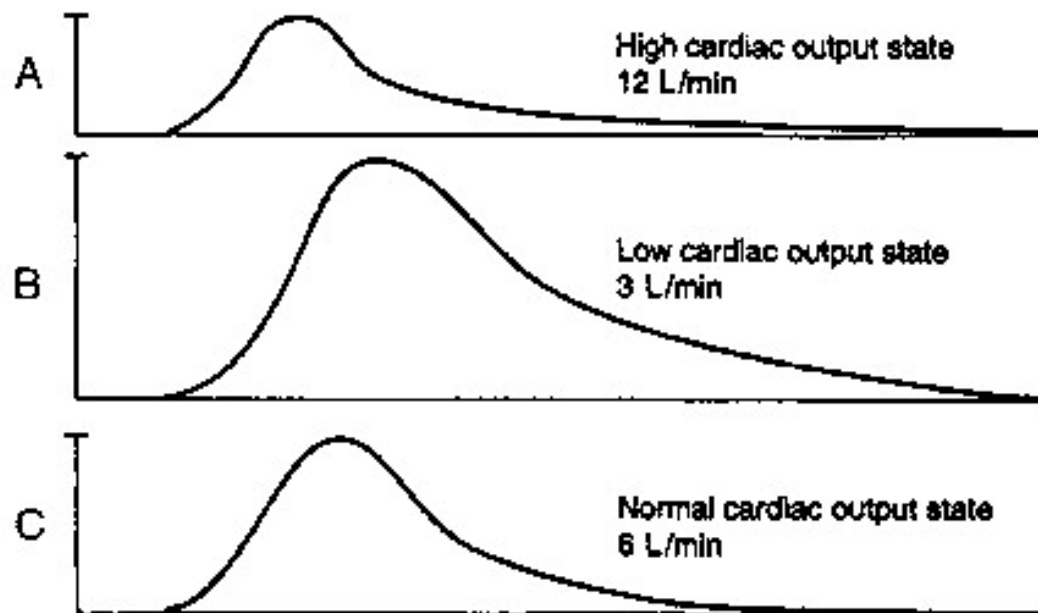
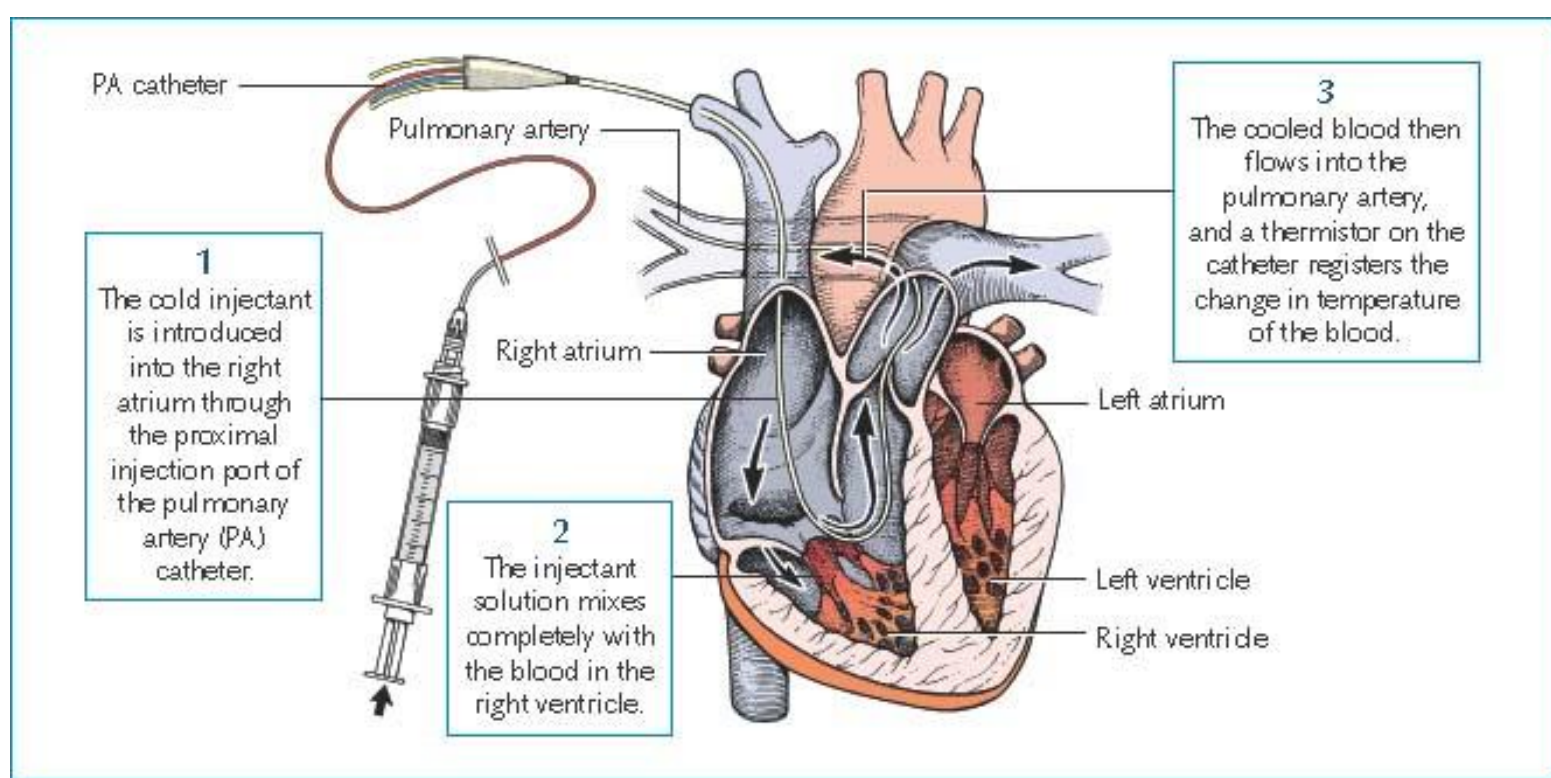


Internal jugular vein

The pulmonary artery catheter



William Ganz and H.J.C. Swan

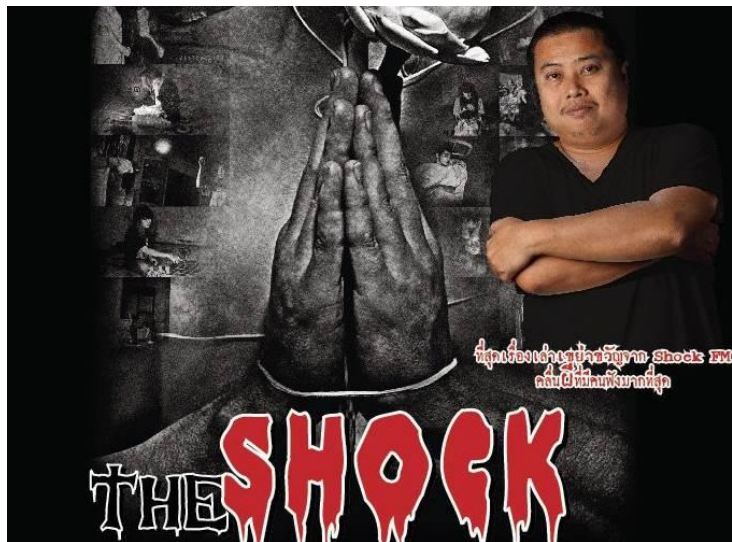


**Thermodilution
Cardiac Output**

Normal Hemodynamic Values

Parameters	Normal values
Central venous pressure (CVP)	< 6 mmHg
Pulmonary capillary wedge pressure (PCWP)	< 6-12 mmHg
Cardiac output (CO) [calculated]	4-8 L/min
Systemic vascular resistance (SVR) [calculated]	800-1600 dynes.sec.cm ⁻⁵

$$SVR = \frac{MAP - CVP}{CO} \times 80$$



SHOCK?



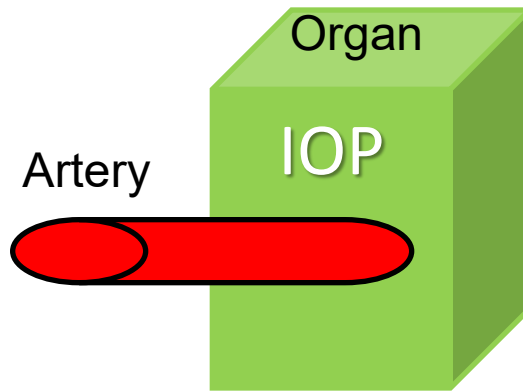
Shock Definition

- **Shock**: a life-threatening acute circulatory failure characterized by decreased tissue perfusion, leading to inadequate oxygen delivery and/or oxygen utilization to meet cellular metabolic demands.
- **Hypotension**: Mean Arterial Pressure* (MAP) < 65 mmHg, SBP < 90 mmHg or ↓ 40 mmHg from baseline
- Hypotension is usually present in shock
- **Hypotension ≠ Shock**

*MAP = (SBP-DBP)/3 + DBP
SBP = systolic blood pressure

Organ perfusion pressure

Organ perfusion pressure = MAP – Intra-organ pressure

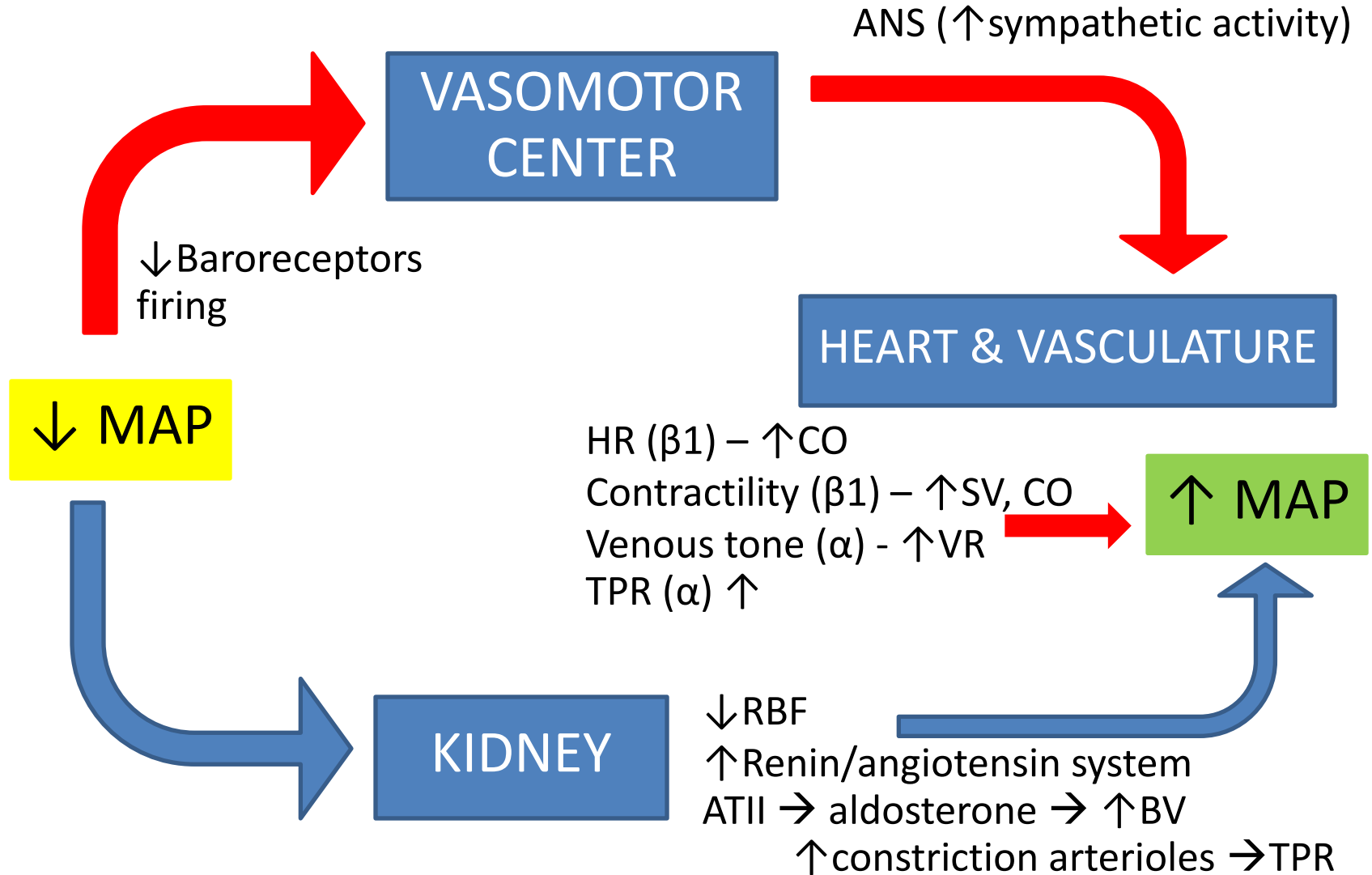


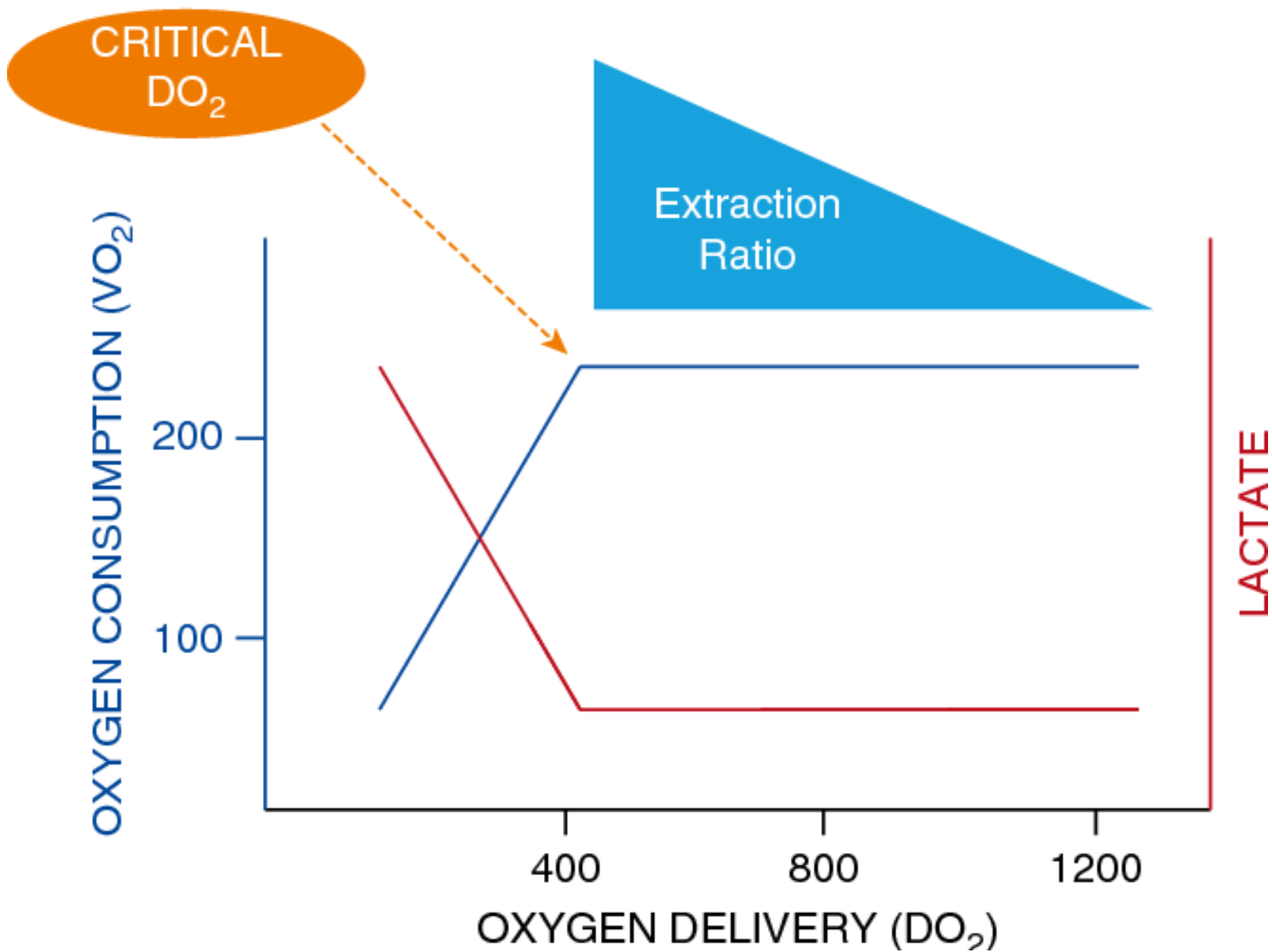
$$CPP = MAP - ICP$$

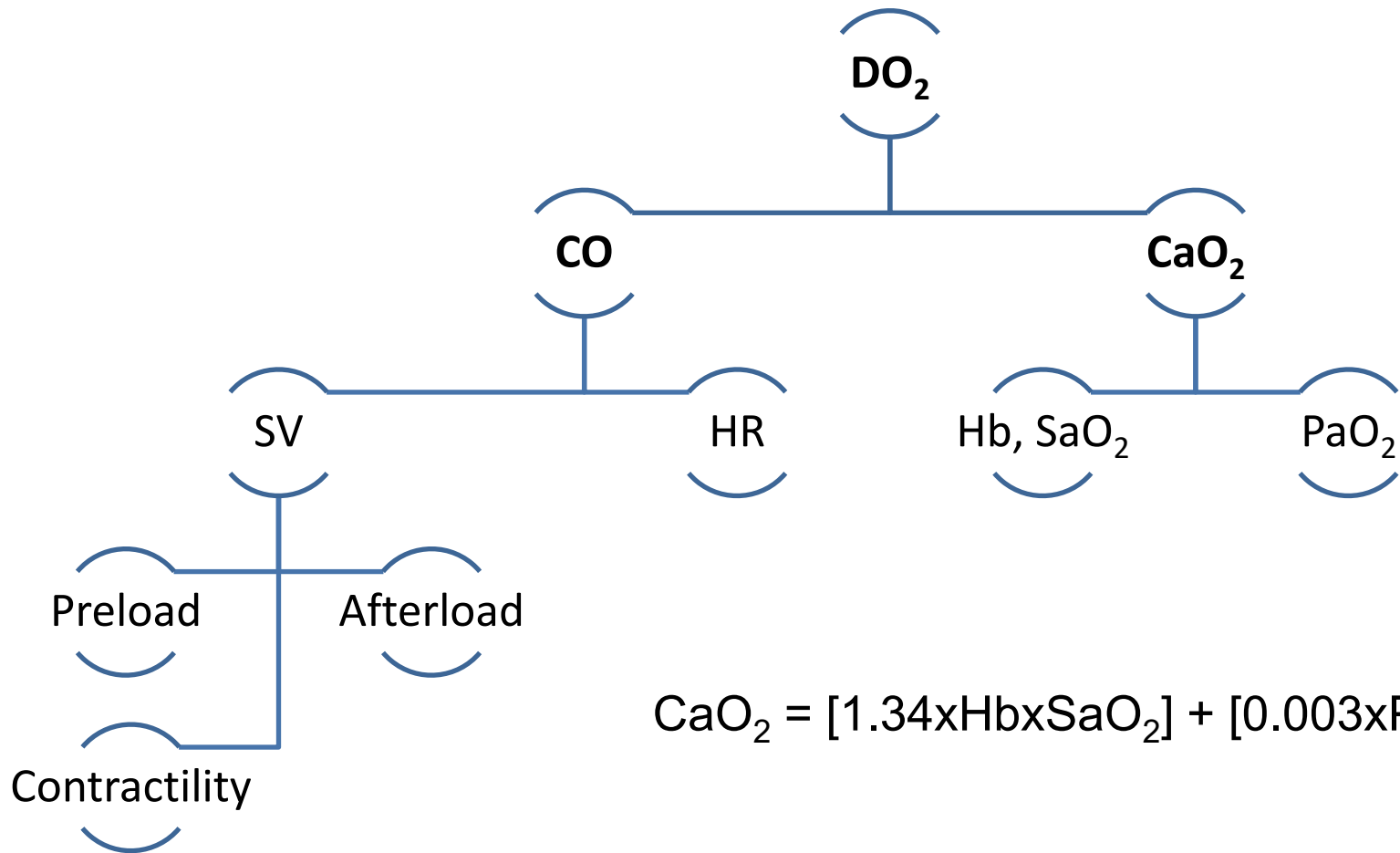
$$55 = 65 - 10$$

CPP, cerebral perfusion pressure
MAP, mean arterial pressure
ICP, intracranial pressure

Regulation of MAP





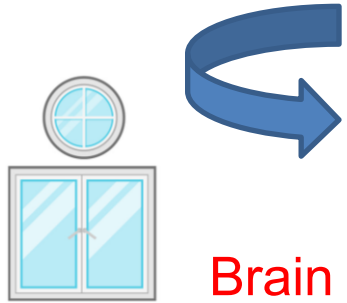


$$CaO_2 = [1.34 \times Hb \times SaO_2] + [0.003 \times PaO_2]$$

DO_2 – oxygen delivery
 CO – Cardiac output
 CaO_2 – Oxygen content
 Hb – Hemoglobin concentration
 SaO_2 – Oxygen saturation
 PaO_2 – Partial pressure of oxygen

Arterial hypotension

Shock



Signs of tissue hypoperfusion



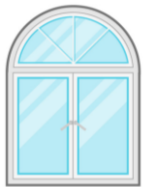
Brain

Altered mental status



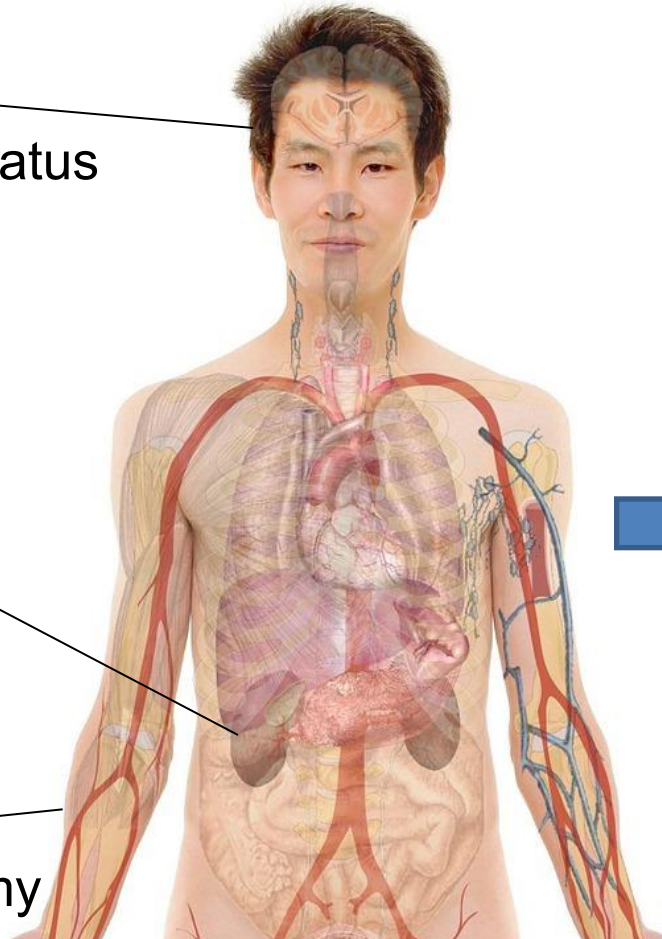
Kidney

Oliguria



Skin

Mottled, clammy



Tachycardia
↑ Blood lactate

Shock 4 types

- Hypovolemic shock
- Cardiogenic shock
- Obstructive shock
- Distributive shock

Shock is a state of acute circulatory failure that can be broadly attributed to four basic mechanisms: hypovolemic, cardiogenic, obstructive, or distributive, or a combination of these

Hemodynamic formula

- $MAP = CO \times SVR$
 $= SV \times HR \times SVR$
 $= (EDV - ESV) \times HR \times SVR$
 $= \frac{EDV}{EDV} \times (EDV - ESV) \times HR \times SVR$
 $= EDV \times EF \times HR \times SVR$



$$BP = \text{preload} \times \text{contractility} \times HR \times \text{afterload}$$

hypovolemic

cardiogenic/
obstructive

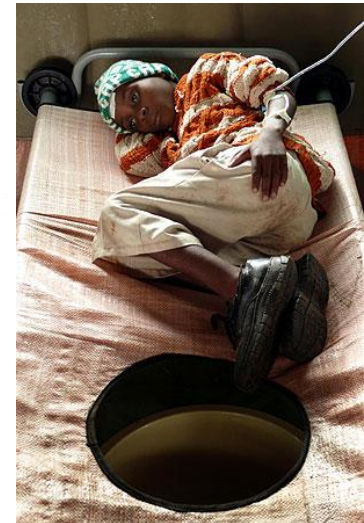
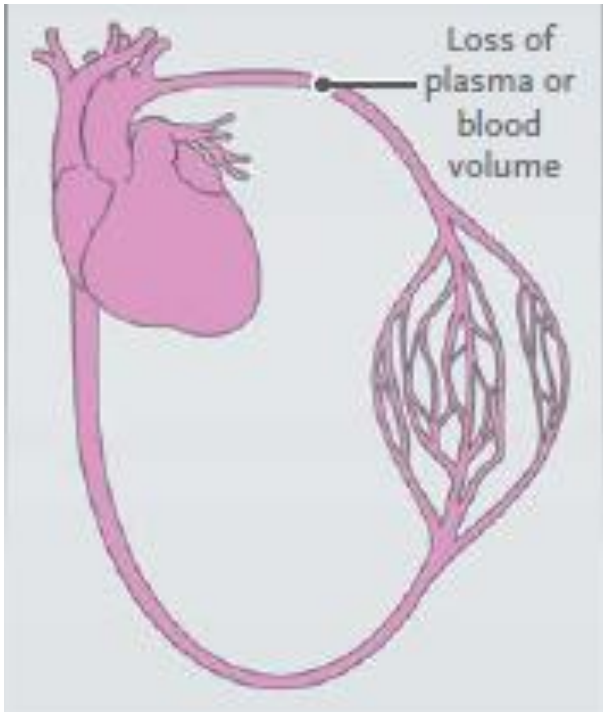
distributive

Hypovolemic shock

$$\text{MAP} - \text{CVP} = \text{EDV} \times \text{EF} \times \text{HR} \times \text{afterload}$$

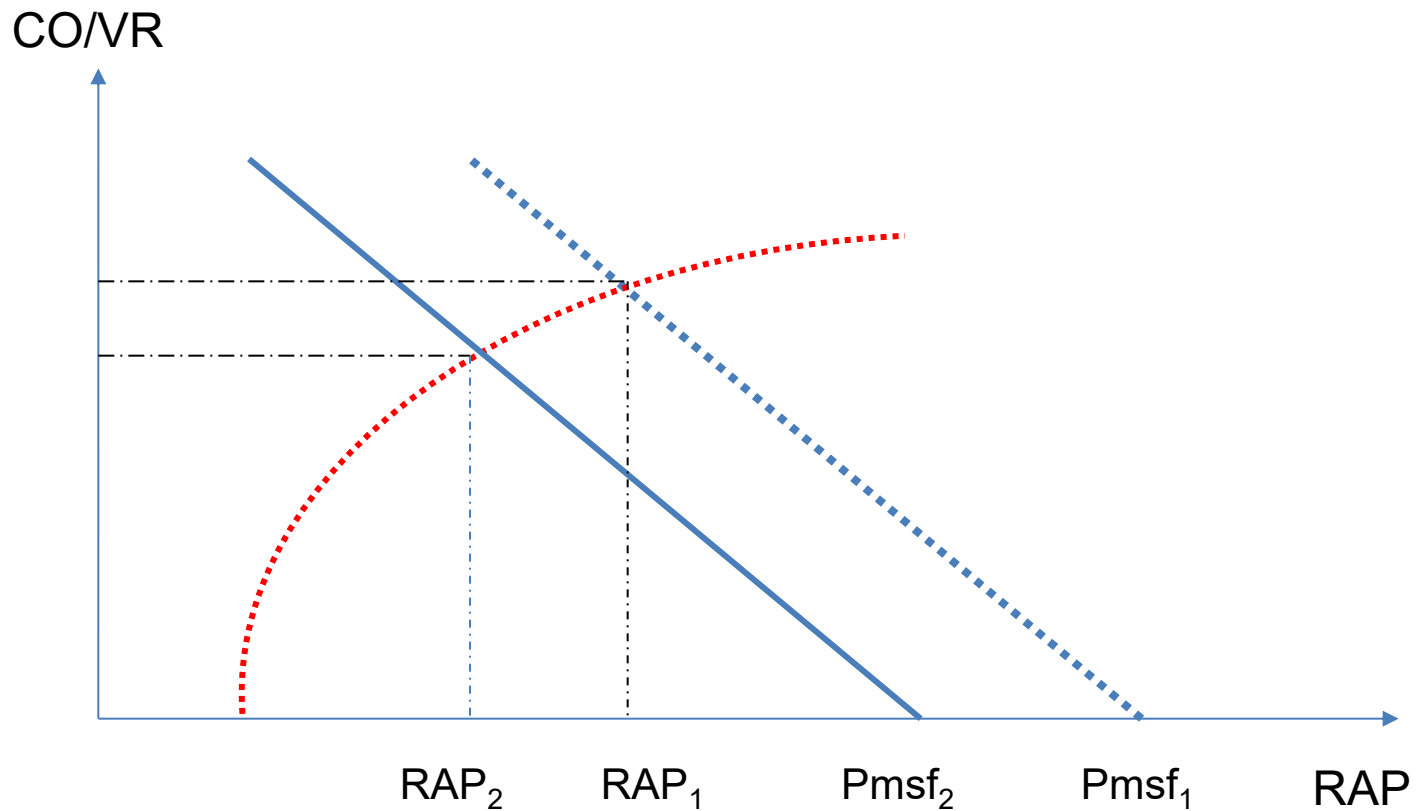
Primary

Compensate



Example causes
Massive hemorrhage
Severe diarrhea

Venous return determinants during hypovolemic shock

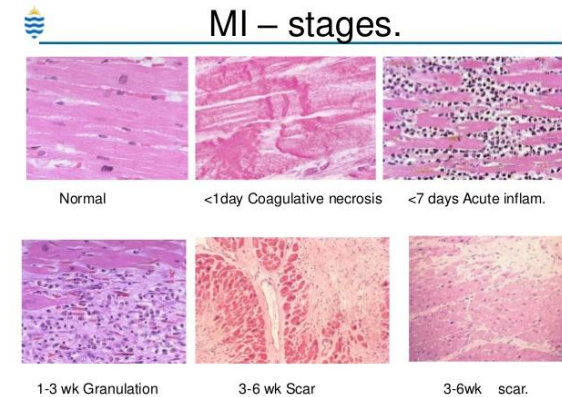
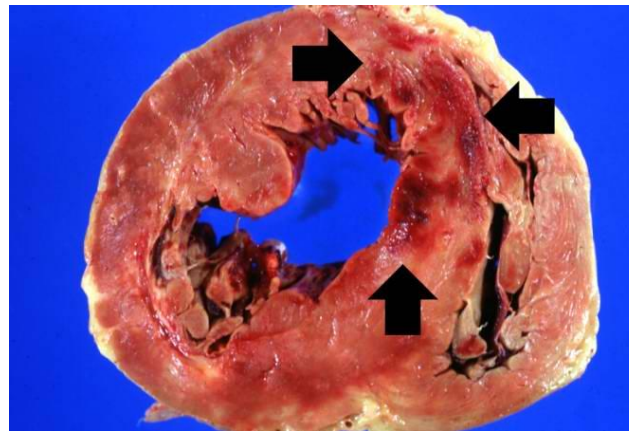
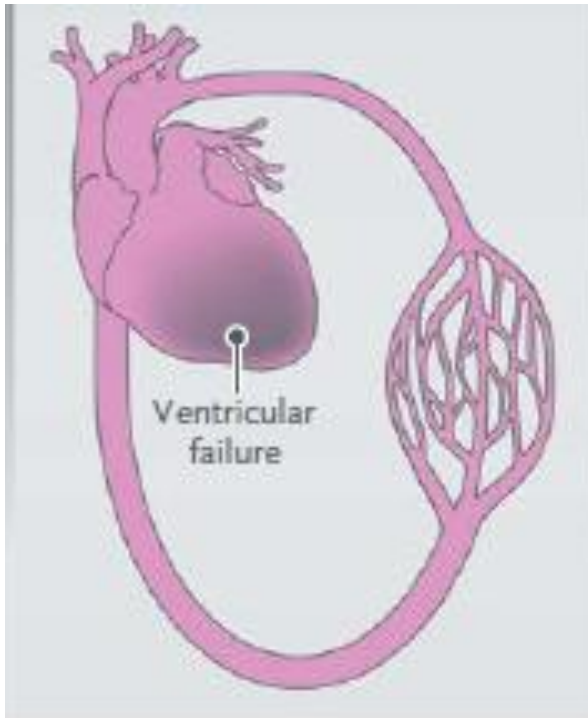


Cardiogenic shock

$$\text{MAP} - \text{CVP} = \text{EDV} \times \text{EF} \times \text{HR} \times \text{afterload}$$

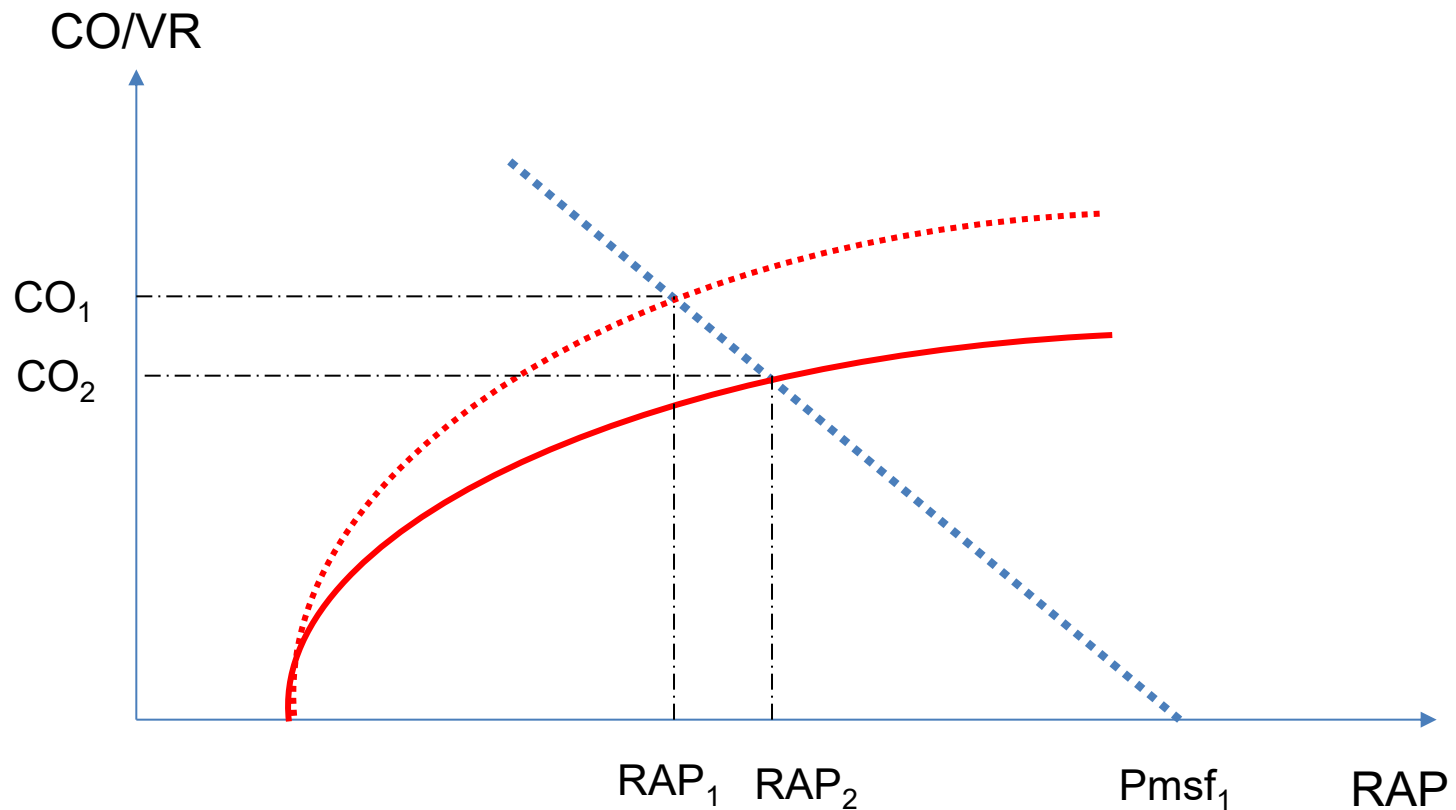
Primary

Compensate



Example causes
Acute myocardial infarction
Myocarditis

Venous return determinants during cardiogenic shock

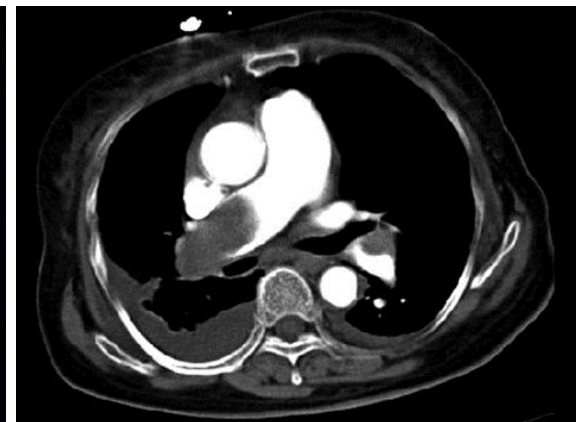
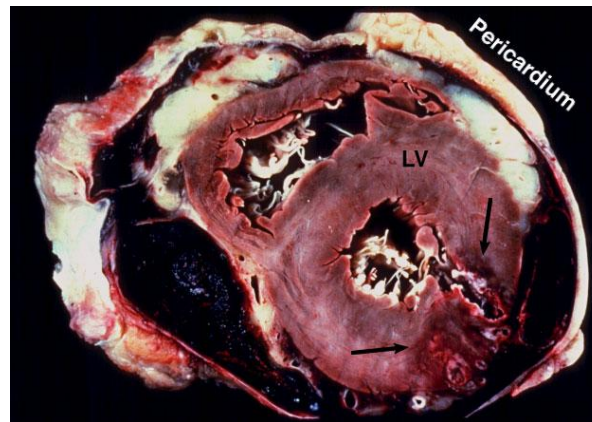
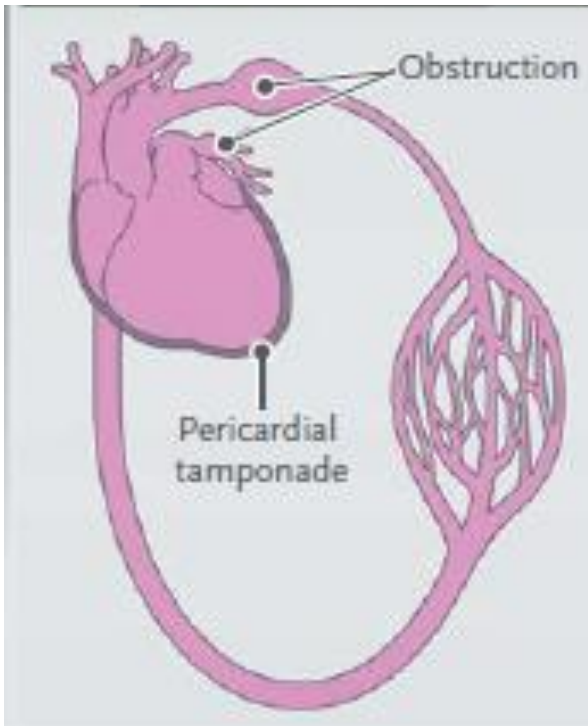


Obstructive shock

$$\text{MAP} - \text{CVP} = \text{EDV} \times \text{EF} \times \text{HR} \times \text{afterload}$$

Primary

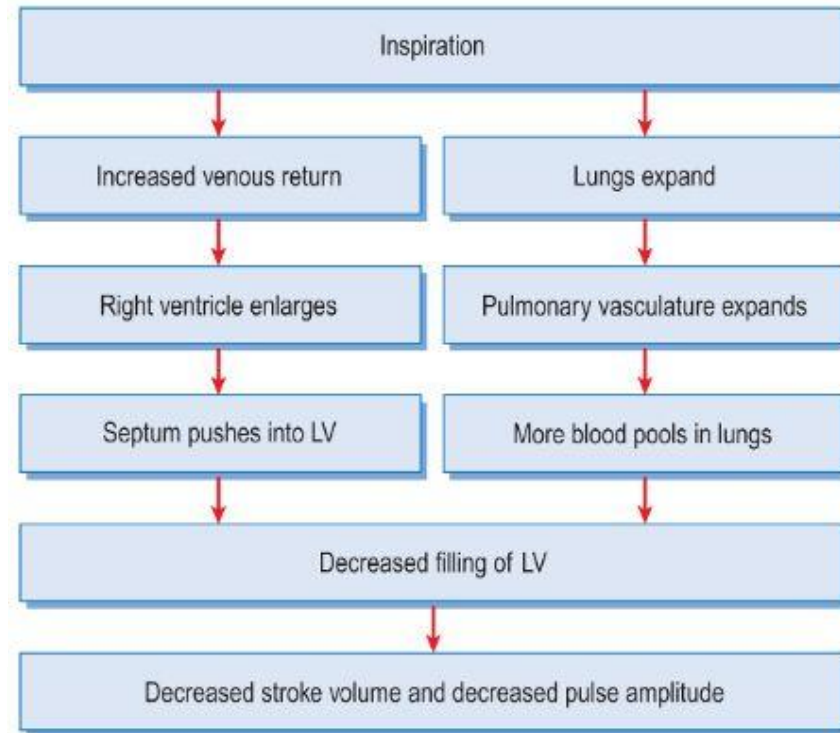
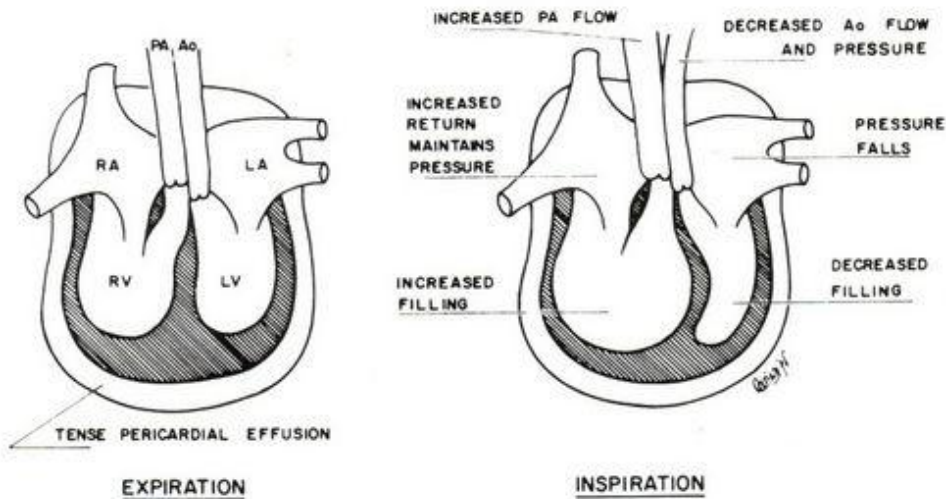
Compensate

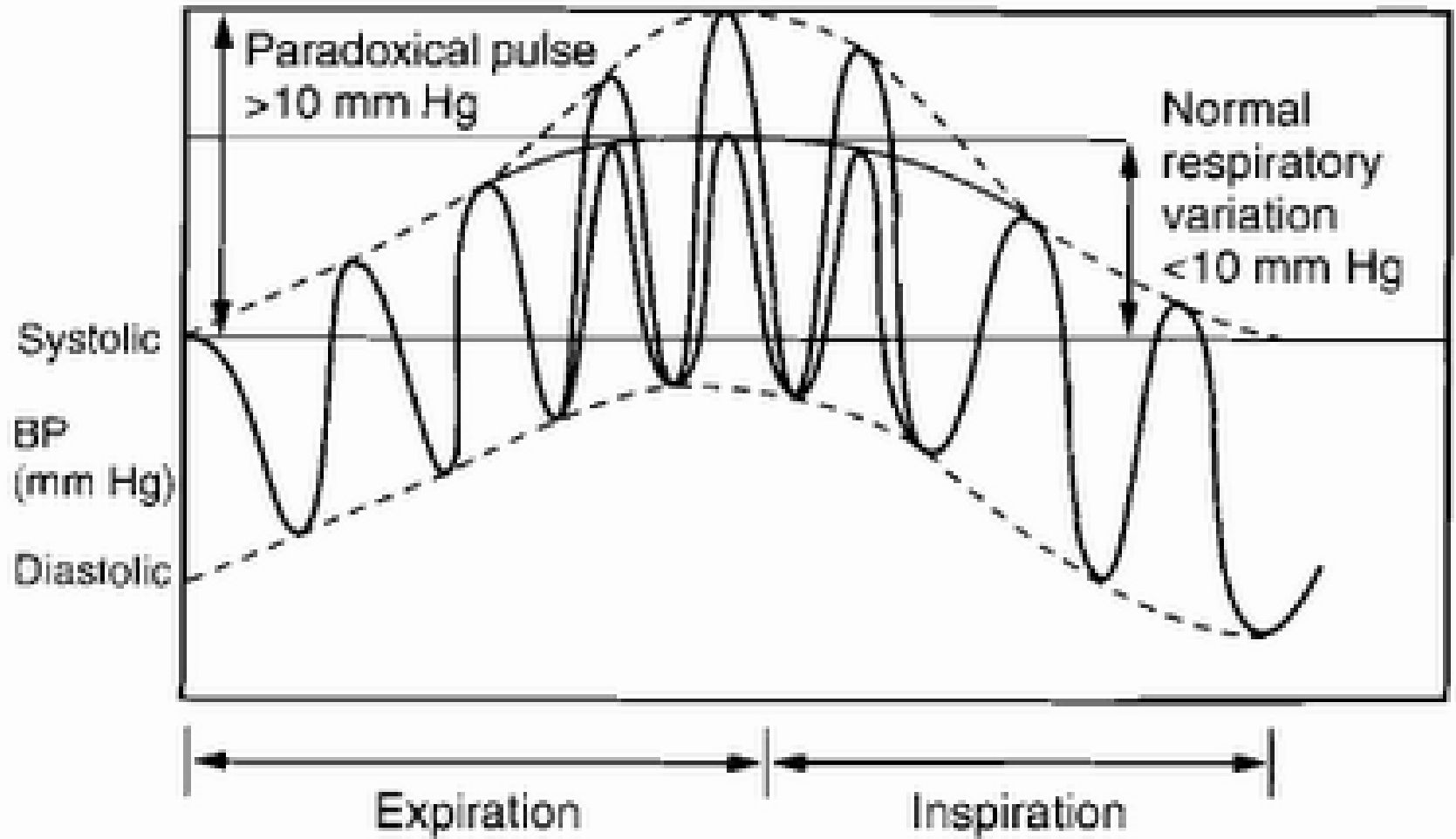


Example causes
Cardiac tamponade
Massive pulmonary embolism

Pulsus paradoxus

MECHANISM OF PULSUS PARADOXUS



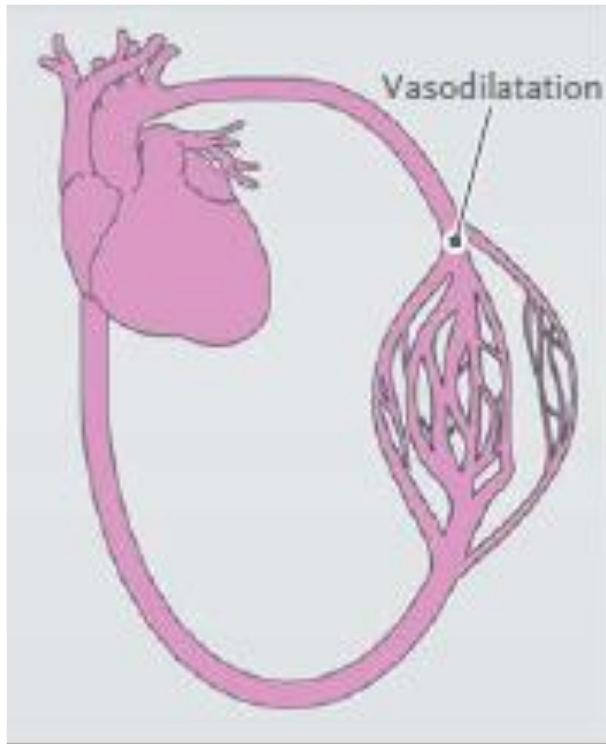


Distributive shock

$$\text{MAP} - \text{CVP} = \text{EDV} \times \text{EF} \times \text{HR} \times \text{afterload}$$

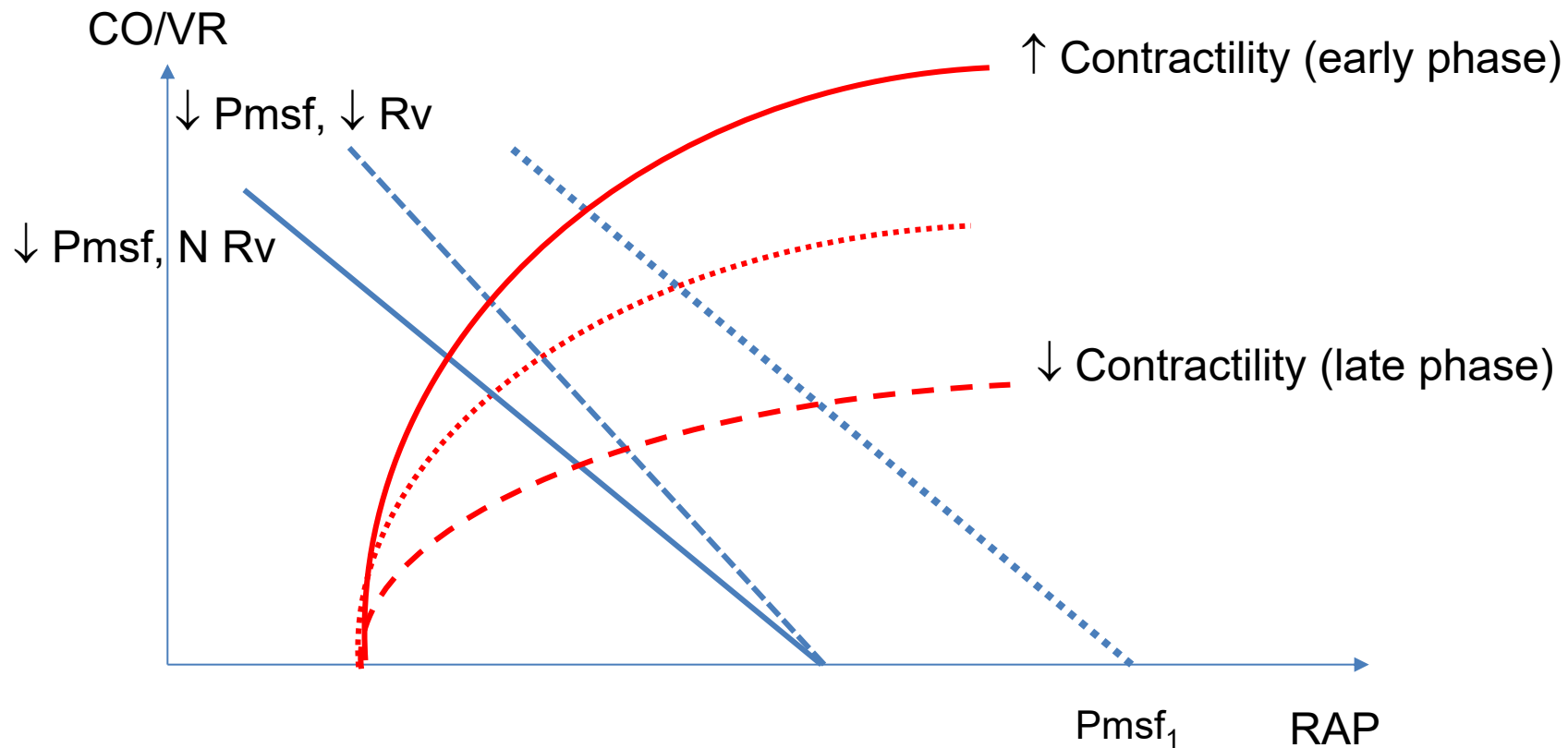
Primary

Compensate



Example causes
Anaphylactic shock
Septic shock

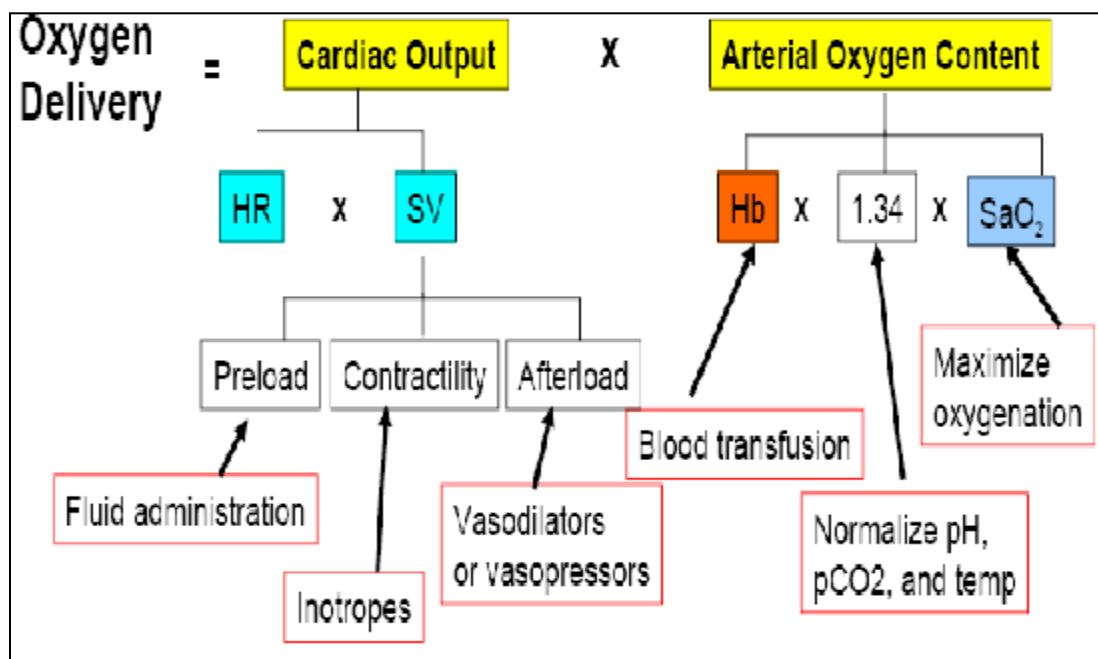
Venous return determinants during septic shock



Hemodynamic Profile in Shock ★

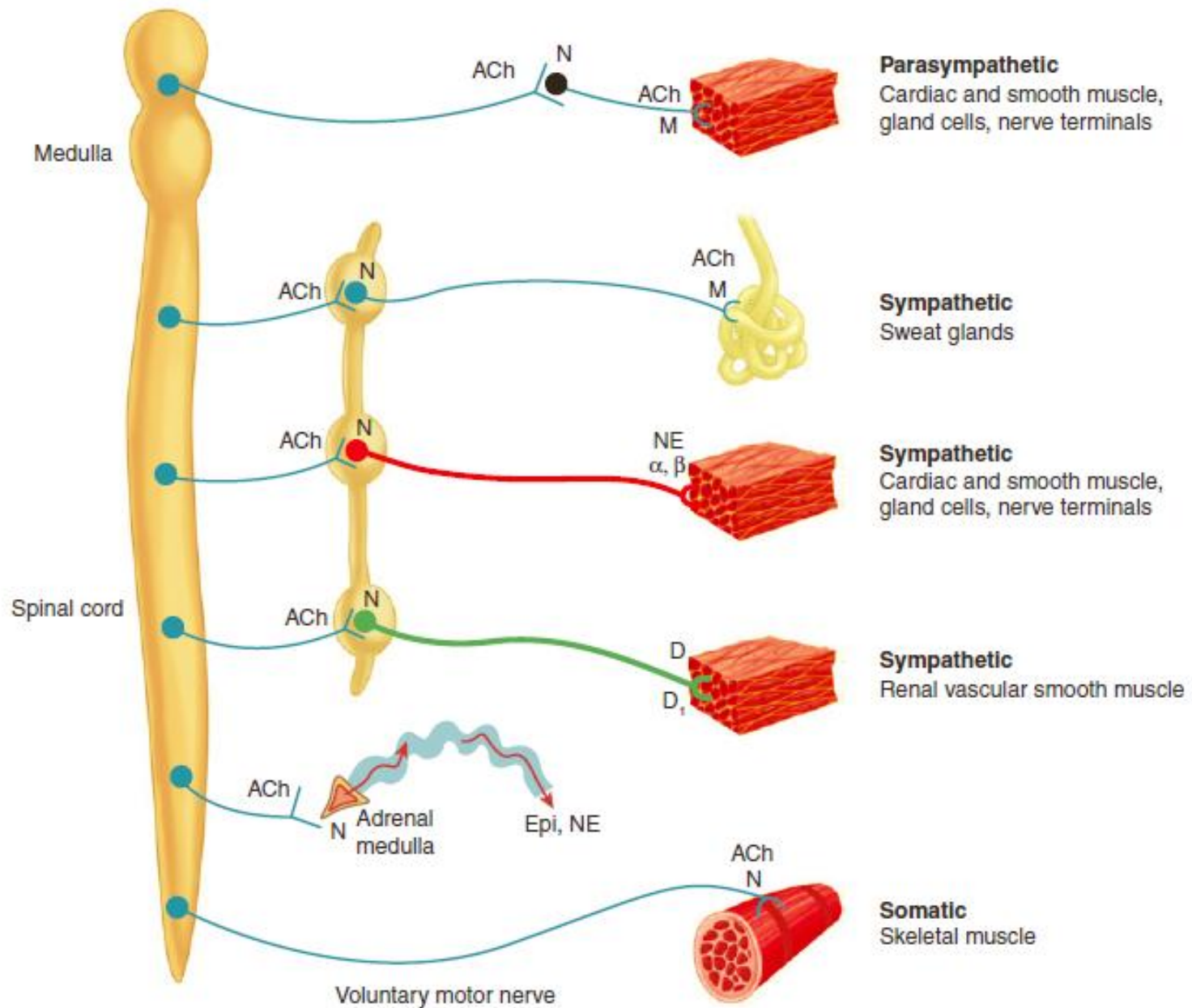
Class of shock	CVP	PAOP	CO/CI	SVR
Hypovolemic	↓	↓	↓	↑
Cardiogenic	↑	↑	↓	↑
Distributive	↓	↓	↑	↓
Obstructive	↑	↓ ↑	↓	↑

Shock resuscitation



VIP rule

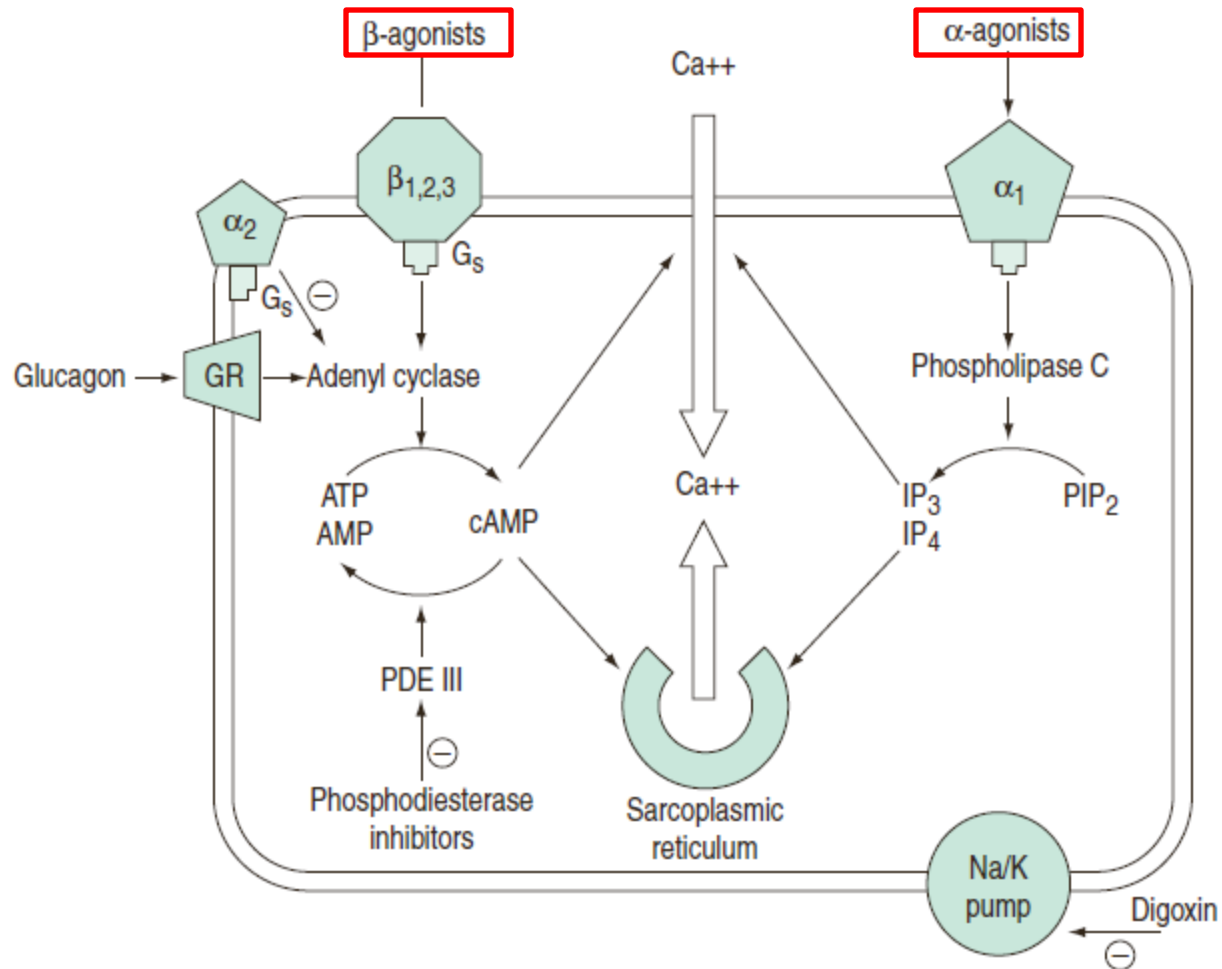
- **V**entilate (oxygen administration)
- **I**nfuse (fluid resuscitation)
- **P**ump (administration of vasoactive agents)



Adrenergic receptor

Alpha receptor	Beta-1 receptor	Beta-2 receptor
• Vasoconstriction • Iris dilatation • Piloerection	• Increase cardiac contractility • Lipolysis	• Vasodilatation • Bronchodilation • Increase glycolysis • Uterine relaxation

Action of vasoactive agents

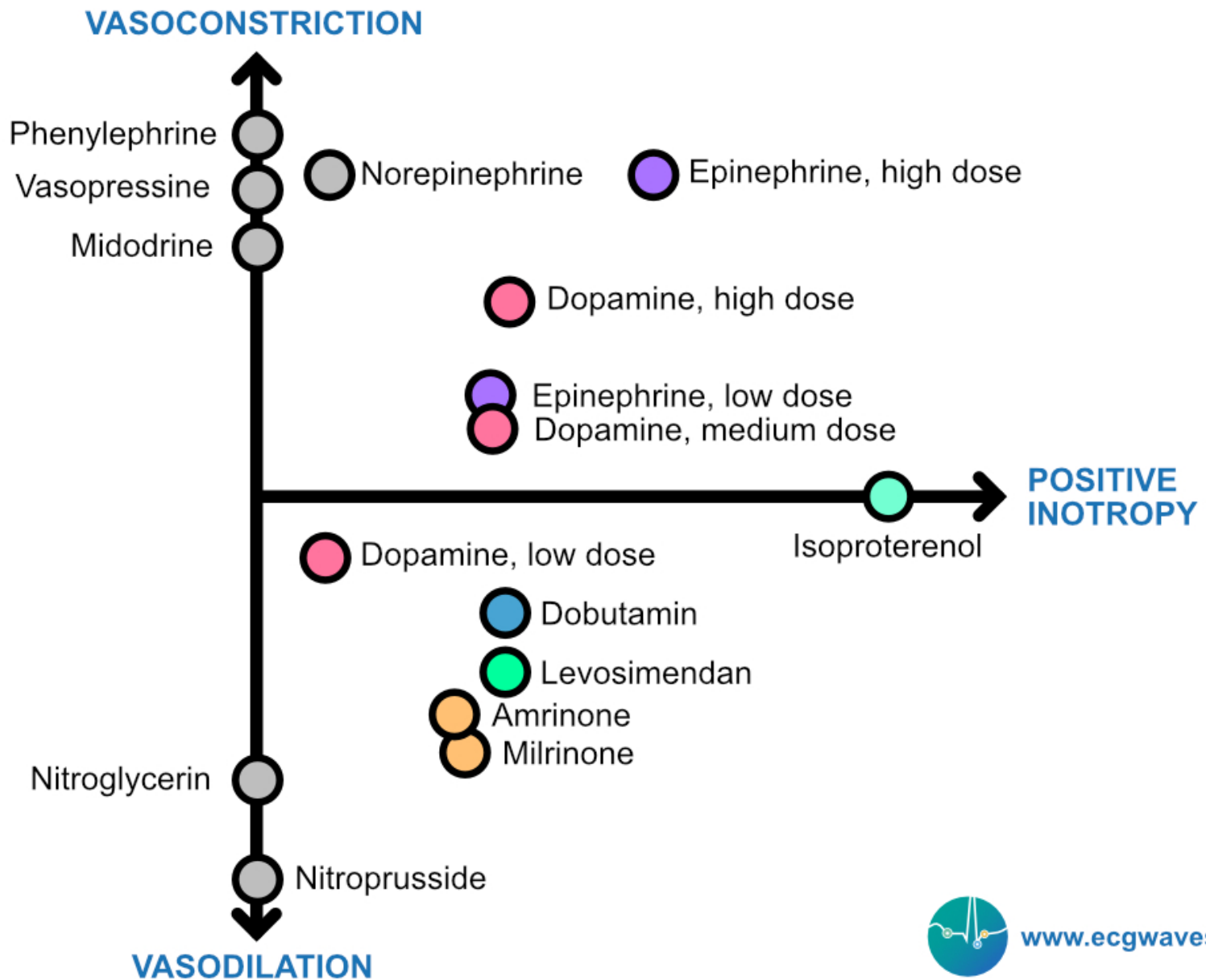


Vasoactive agents

- Positive inotropic effect : contractility
- Positive chronotropic effect : HR
- Vasopressor effect : increase SVR (afterload)

Vasoactive agents

	Alpha	Beta-1	Beta-2	DA
Dobutamine	-	++	+	-
Dopamine (low dose)	-	-	-	++
Dopamine (moderate dose)	-	+++	+++	-
Dopamine (high dose)	++	+++	+++	-
Epinephrine	+++	++++	+++	-
Norepinephrine	+++	+	-	-

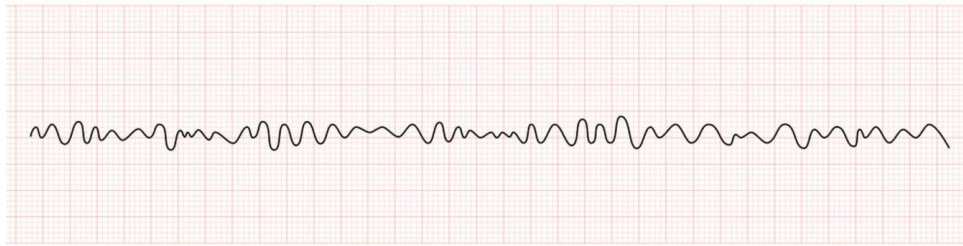




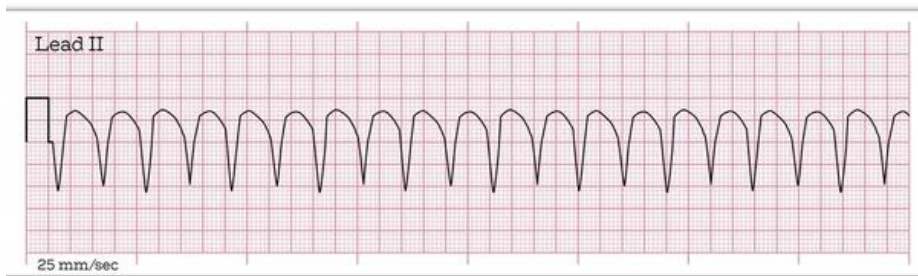
Case study

Cardiac arrest rhythms (no pulse)

Shockable rhythms



Ventricular fibrillation, VF

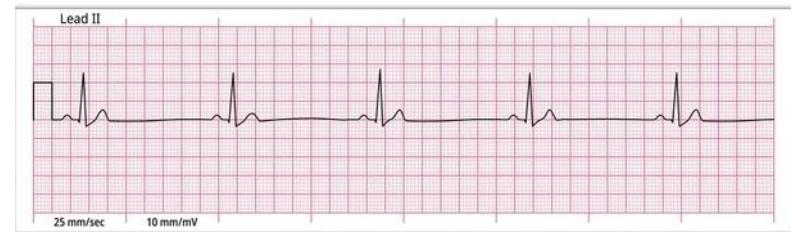


Ventricular tachycardia, VT

Non-shockable rhythms

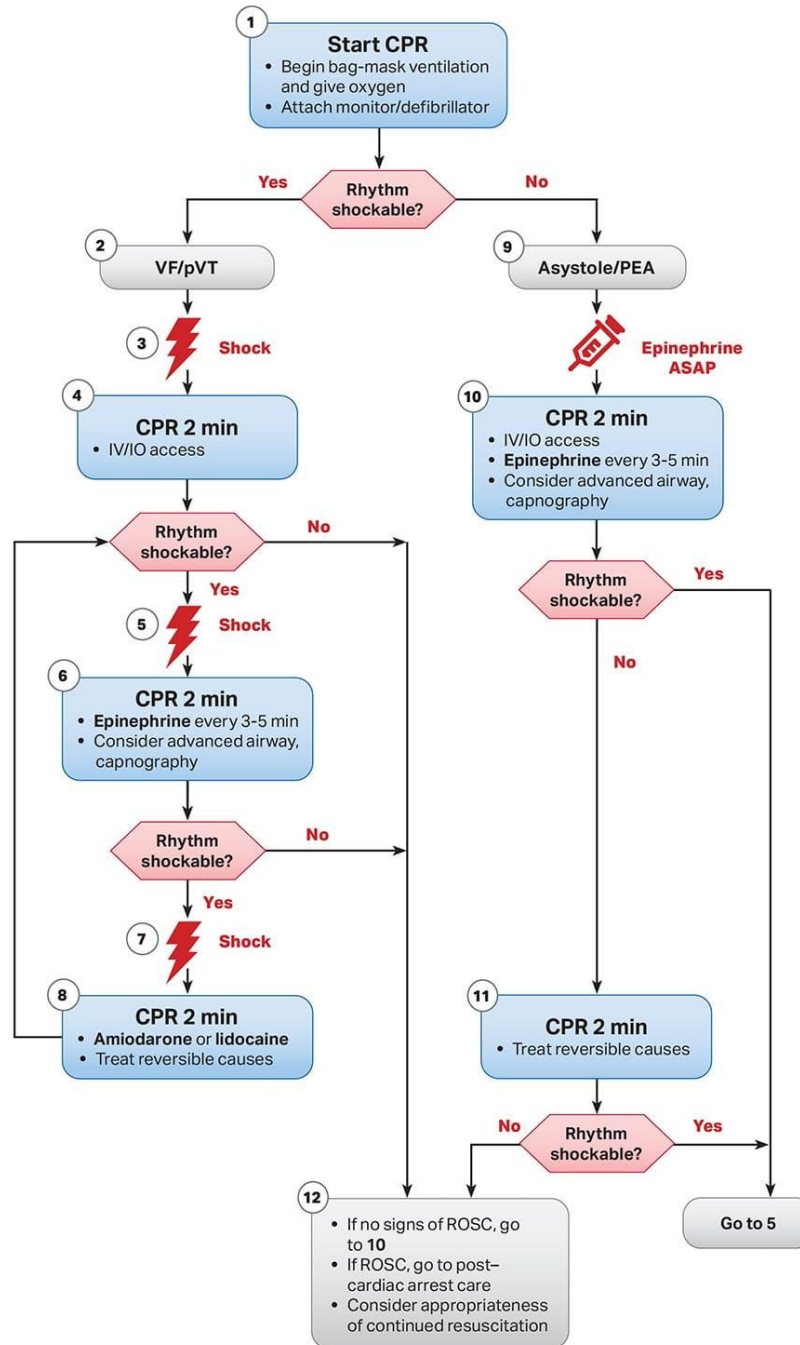


Asystole



Pulseless electrical activity (PEA)

Adult Cardiac Arrest Algorithm (VF/pVT/Asystole/PEA)



High-Quality CPR
- Push hard (at least 2 inches [5 cm]). - Push fast (100-120/min) and allow complete chest recoil. - Minimize interruptions in compressions. - Avoid excessive ventilation. - Change compressor every 2 minutes, or sooner if fatigued. - If no advanced airway, use 30:2 compression-ventilation ratio. - If advanced airway in place, give 1 breath every 6 seconds (10 breaths/min) with continuous chest compressions. - Continuous waveform capnography - If ETCO_2 is low or decreasing, reassess CPR quality.
Shock Energy for Defibrillation
Biphasic: Manufacturer recommendation (eg, initial dose of 120-200 J); if unknown, use maximum available. Second and subsequent doses should be equivalent, and higher doses may be considered. Monophasic: 360 J
Drug Therapy
Epinephrine IV/IO dose: 1 mg every 3-5 minutes Amiodarone IV/IO dose: First dose: 300 mg bolus Second dose: 150 mg or Lidocaine IV/IO dose: First dose: 1-1.5 mg/kg Second dose: 0.5-0.75 mg/kg
Advanced Airway
- ET intubation or supraglottic advanced airway - Continuous waveform capnography or capnometry to confirm and monitor ET tube placement
Reversible Causes
- Hypovolemia - Hypoxia - Hydrogen ion (acidosis) - Hypo-/hyperkalemia - Hypothermia - Tension pneumothorax - Tamponade, cardiac - Toxins - Thrombosis, pulmonary - Thrombosis, coronary

Case study

- Mr. Harry, a 66-year-old, T2DM, HTN, chronic stable angina, recent admitted with congestive heart failure.
- He was sent to the ED due to nausea, chest pain and numbness on his arm.

ACLS EMERGENCY DRUGS

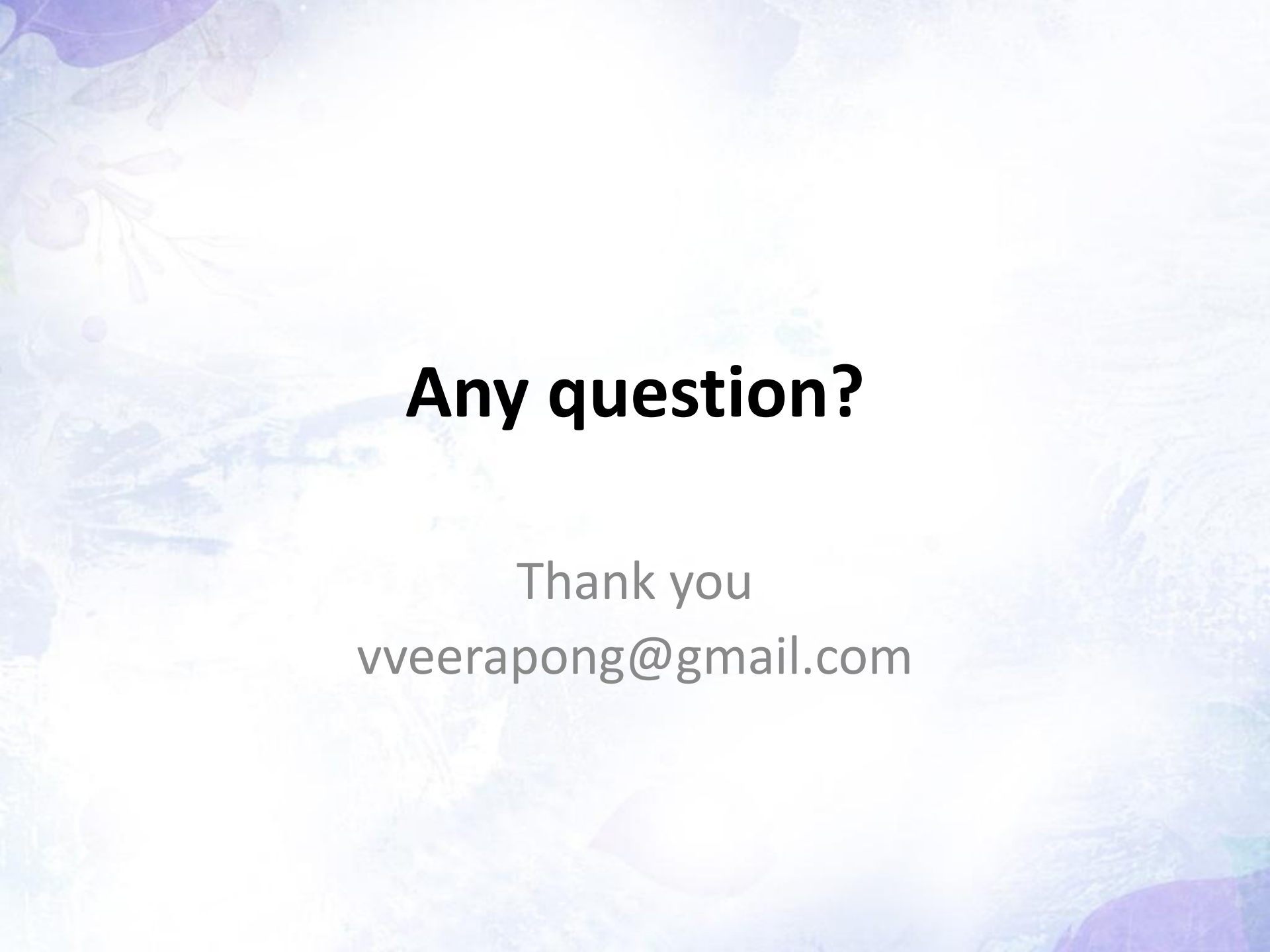


DRUG	INDICATION	TYPICAL DOSE
 EPINEPHRINE (ADRENALINE)	• Cardiac arrest (PEA/Asystole) • Anaphylaxis	1 mg IV/IO every 3–5 min
 AMIODARONE	• Ventricular fibrillation (VF) • Pulseless VT	300 mg IV/IO bolus then 150 mg IV/IO
 LIDOCAINE	• Pulseless VT • PVCs with pulse	1–1.5 mg/kg IV bolus then 0.5–0.75 mg/kg (repeat as needed)
 ATROPINE	• Symptomatic bradycardia	1 mg IV every 3–5 min (max 3 mg)
 ADENOSINE	• Stable SVT (narrow regular) • Diagnostic (vagal blockade)	6 mg rapid IV push then 12 mg (if needed)



Assignment สำหรับคะแนน process

- ตอบคำถามทาง **Google form** ด้วยตนเอง เพื่อทบทวนความเข้าใจ
- ส่งคำตอบตามเวลา = ได้คะแนนเต็ม
- หหมดเขตรับคำตอบ **วันศุกร์ ที่ 18 กันยายน 2569 เวลา 16.00 น.**
- ไม่เปิดรับคำตอบหลังจากหมดเขต



Any question?

Thank you
vveerapong@gmail.com